

REVIEW ARTICLE OPEN ACCESS

A Systematic Scoping Review of Biogas as an Instrumental Technology for a Circular Economy and a Way Forward to Ensure Sustainable Development in the Context of Global South Nations

Tewodros Tarekegn Lapiso  | Hynek Roubík 

Department of Sustainable Technologies, Faculty of Tropical AgriSciences, Czech University of Life Sciences Prague, Prague, Czech Republic

Correspondence: Hynek Roubík (roubik@ftz.czu.cz)

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ABSTRACT

Sustainable development (SD) was introduced in 1987 by the Brundtland Commission report. This leads to the induction of the circular economy (CE). Within these two frameworks, biogas technology plays a crucial role in facilitating the transformation of organic waste into energy and organic fertilizer without producing non-recyclable waste. This study aims to elucidate how peer-reviewed research articles employed research methodologies to investigate the interconnections between biogas, CE, and SD within the Global South context. A systematic scoping review was performed with strong adherence to the PRISMA protocol. Therefore, we applied two analytical approaches: a manual interpretative coding analysis and a bibliometric analysis utilizing RStudio's Biblioshiny interface. The results indicate the dominance of quantitative methodology, with the prevalence of experimental and survey research designs. The direct interconnection of biogas, CE, and SD is hard to find in the inquiries of the reviewed articles. Moreover, CE has the least thematic representation compared to the combination of SD and CE as a distinct theme alongside SD. The publication of peer-reviewed research articles exhibits an increasing trend, with China occupying a prominent position. This review substantiates the instrumental role and status of Biogas in the effective establishment of the CE and SD in Global South Nations. Furthermore, we recommend that future studies should explicitly examine biogas in the framework of CE and SD. Policies should be tailored to the socio-economic contexts of Global South Nations through the application of instruments like investment support through subsidies or loans, and incentivizing research and development efforts.

1 | Introduction

SD, a comprehensive developmental politics term, was first introduced in the Brundtland Commission report in 1987 (World Commission on Environment and Development 1987). It laid the groundwork for a global trend focusing on rights over needs (Redclift 2005). In 1992, the Rio Earth Summit

forged a global consensus on SD (United Nations Division for Sustainable Development 1992). The release of Millennium Development Goals (MDGs) at the 2000 Millennium Summit exemplified this consensus. By 2012, the consensus further strengthened through the formulation of the Sustainable Development Goals (SDGs) at the Rio+ Conference (United Nations—General Assembly 2000; United Nations 2012). In

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September 2015, the United Nations General Assembly adopted SDGs with 17 goals, 169 targets, and a total of 248 indicators, of which 231 are unique, effective in 2016 following the MDGs phaseout (United Nations—General Assembly 2000; United Nations 2015).

Importantly, SD underscores the necessity for a harmonious integration of economic development, social inclusion, and environmental preservation, thus facilitating the development of innovative solutions that enhance resilience and sustainability across various sectors (Bhandari 2019; Ogryzek 2023). These solutions not only aim to meet the needs of the present but also ensure that future generations can thrive in a balanced and equitable world (World Commission on Environment and Development 1987). Such an approach gives rise to the notion of the CE (Murray et al. 2017). CE has a protracted history, making tracing its exact inception difficult. Nonetheless, the European Union (EU), through its Directive 2008/98/EC on waste, alongside China's CE promotion law, significantly increased public awareness, thereby contextualizing SD's application (Murray et al. 2017; Suárez-Eiroa et al. 2019). Consequently, this concept garnered attention from scholars investigating it as a plausible method for attaining SD and establishing a novel economic model (Geissdoerfer et al. 2017; Homrich et al. 2018).

Consistently, CE is committed to producing products such as renewable energy and others from waste materials within a resource-circular closed-loop system (Ghazanfari 2023; Ghisellini et al. 2016). It emphasizes the interconnections among water, food, waste, and energy (El-Ramady et al. 2022; Lin et al. 2021; Siaw et al. 2022). This is evidenced in a life cycle assessment of biogas in Brazil's cassava industry (Lin et al. 2021) and the biogas potential of agricultural waste

(Kapoor et al. 2020). In this context, biogas emerges to play a crucial role in a circular economy, converting organic waste without creating non-recyclable byproducts (Burg et al. 2023; D'Adamo et al. 2023; Reynolds et al. 2022).

Hence, biogas technology uses naturally occurring microorganisms for anaerobic digestion to convert organic waste into renewable energy and organic fertilizer (Bond and Templeton 2011; D'Aquino et al. 2022). The gas consists of 50%–70% methane (CH_4) and 30%–50% carbon dioxide (CO_2), with trace gases (Bond and Templeton 2011). It is scalable, using local resources and advanced methods for better productivity and safety (Kougias and Angelidaki 2018). Biogas is ideal for Global South Nations, which are less economically developed (developing) and mostly in the southern hemisphere (Li, Gan, et al. 2023; Wang et al. 2020; Kowalski 2021). With increased waste from population growth and urbanization, averaging 0.56 kg/person/day, mainly organic and recyclable (52% and 26%), these regions can become green energy and job hubs within the CE and SD frameworks. Challenges include poor waste management, underused biogas facilities, and financial, infrastructural, regulatory, and expertise deficiencies (Bundhoo 2018; Ebhota and Tabakov 2024). These scenarios are further illustrated in detail in Figure 1.

Aligning with this, the study seeks to elucidate how peer-reviewed research articles address the interconnections between biogas technology, CE, and SD within the Global South Nations. Furthermore, it aims to evaluate the methodological approaches and designs present in the reviewed articles, examine themes that emerge as sufficiently prominent to be acknowledged within the reviewed articles, and identify and highlight the existing gaps in the application of methodologies

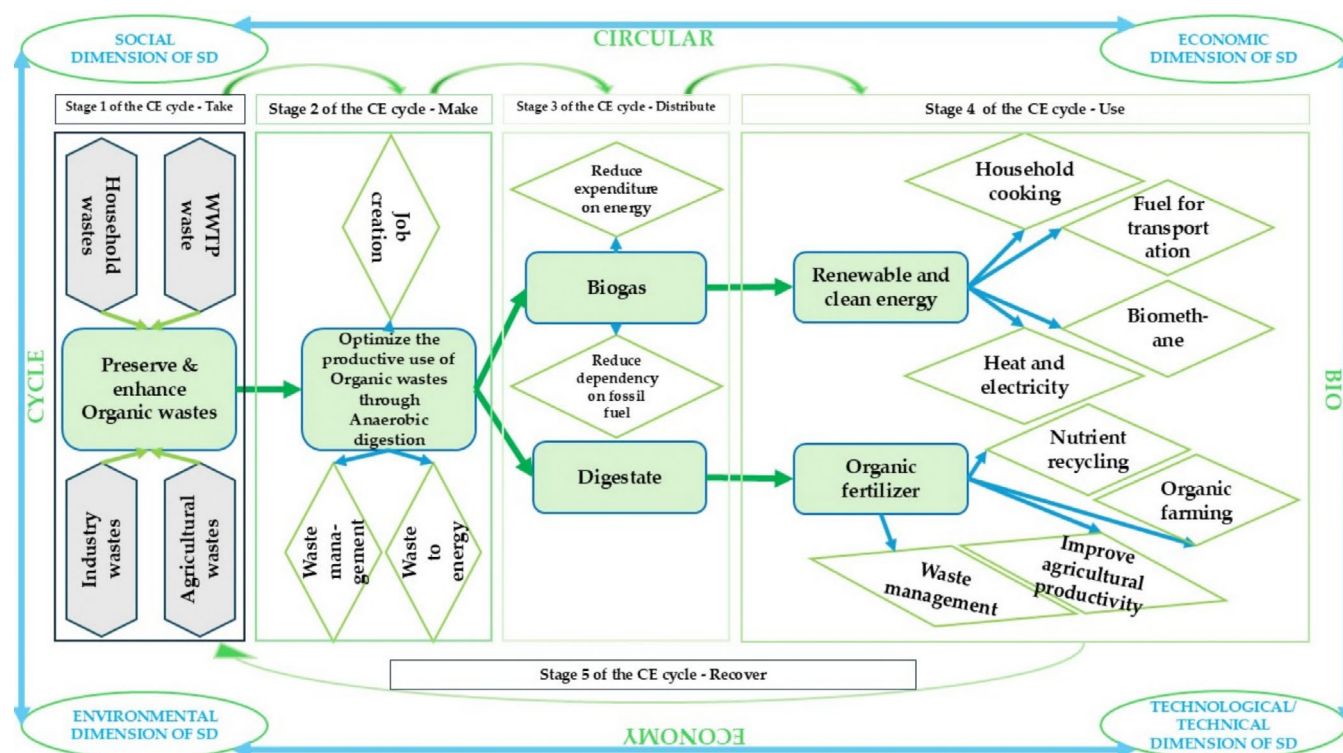


FIGURE 1 | Theoretical/empirical framework of the study that links biogas technology, CE, and SD. Source: Author.

and the discussion of themes. Upon these objectives, this review addresses questions like—in what manner did the reviewed research articles employ research methods in addressing their inquiries? And what are the principal themes of the reviewed studies in associating biogas technology with CE and SD within the context of Global South Nations?

Therefore, this underscores the need to integrate biogas technology and CE to demonstrate their role in achieving SD. The article reviews peer-reviewed articles linking biogas, CE, and SD in Global South Nations, aiming to show how these nations can implement biogas technology to promote a CE that supports SD. The review suggests an evidence-based approach to create a structured knowledge base for optimizing limited resources, addressing the need for scholarly study in Global South Nations, and offering guidance on biogas, CE, and SD relationships. Accordingly, this review article is organized into five chapters: Chapter 2 details the methodology, Chapter 3 interprets the results with supporting figures and charts, Chapter 4 discusses findings in context, and Chapter 5 highlights the study's significance and offers recommendations for decision-makers and future research.

2 | Method

The study employs a systematic scoping review to evaluate the breadth and extent of included studies, analyzing utilized research methodologies and thematic considerations to explicitly address the research question (Munn et al. 2018). The detailed procedural applications of the research design are explicated in the following sections.

2.1 | Keywords, Searched Databases, Search Strategy, and Procedure

As demonstrated in Figure 2, all search strategies, procedures, and the selection of keywords as well as databases are executed in accordance with the standards established by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 protocol (checklist and diagram) (Page et al. 2021).

Subsequently, as Peters et al. (2015) revealed, the search strategy development is structured through three integral steps, regarded as the benchmark for systematic reviews. Initially, a comprehensive search is executed across six databases: Scopus, Google Scholar, ScienceDirect, Wiley, Springer, and Web of Science. The obtained search results undergo a thorough evaluation of the availability and quality of resources, achieved through examining article titles, abstracts, and keyword analysis. During this procedure, all potential synonym keywords for subsequent steps are compiled, and a decision is made for the selection of Web of Science and Scopus as the preferred databases. Moreover, this step facilitates the decision to employ a systematic scoping review as a research design deemed suitably beneficial to this study based on the availed resources at hand.

Afterward, the search strategy box in Figure 2 shows the next stage, which involved a final search in Web of Science and

Scopus using two search methods: combining keywords with synonyms. The first method links biogas keywords with SD and Global South Nations {e.g. (Biogas OR Biofuel OR Biomethane OR “Anaerobic Digestion” OR Bioenergy) AND (“Sustainable Development” OR “Sustainable development goal*” OR SDG* OR Sustainability) AND (“Global South Nation*” OR “Developing Nation*” OR “Africa* countr*” OR “Latin America countr*” OR “Southeast Asia countr*”)}, while the second pairs biogas with the circular economy and Global South Nations {e.g. (Biogas OR Biofuel OR Biomethane OR “Anaerobic Digestion” OR Bioenergy) AND (“Circular Economy” OR “Circular Bioeconomy” OR Circularity) AND (“Global South Nation*” OR “Developing Nation*” OR “Africa* countr*” OR “Latin America countr*” OR “Southeast Asia countr*”)}. The approach was consistent in both databases, with minor user interface adjustments. The final stage required refining results using inclusion–exclusion criteria.

2.2 | Inclusion and Exclusion Criteria

The refinement process was administered according to a pre-defined set of inclusion and exclusion criteria. The criteria employed are listed as follows:

Inclusion:

- Type of paper: Only peer-reviewed research articles
- Language: only articles written in English
- Period: January 1988 to August 2023
- Geographic coverage: The Global South Nations
- Areas: papers on anaerobic digestion/biogas with either sustainable development or circular economy, and related ones
- Accessibility: Only fully accessible articles

Exclusion:

- Non-peer-reviewed articles.
- Publications not presented in the English language.
- Works published before January 1988 and after August 2023.
- Articles concentrating on developed countries or Global North nations.
- Articles are confined behind a paywall and require a specialized license for access.

The articles chosen based on the defined inclusion and exclusion criteria progressed to the categorization phase in preparation for the analysis.

2.3 | Grouping of Studies for Synthesis

The peer-reviewed research articles deemed suitable for inclusion in the analysis of this study were categorized into two groups:

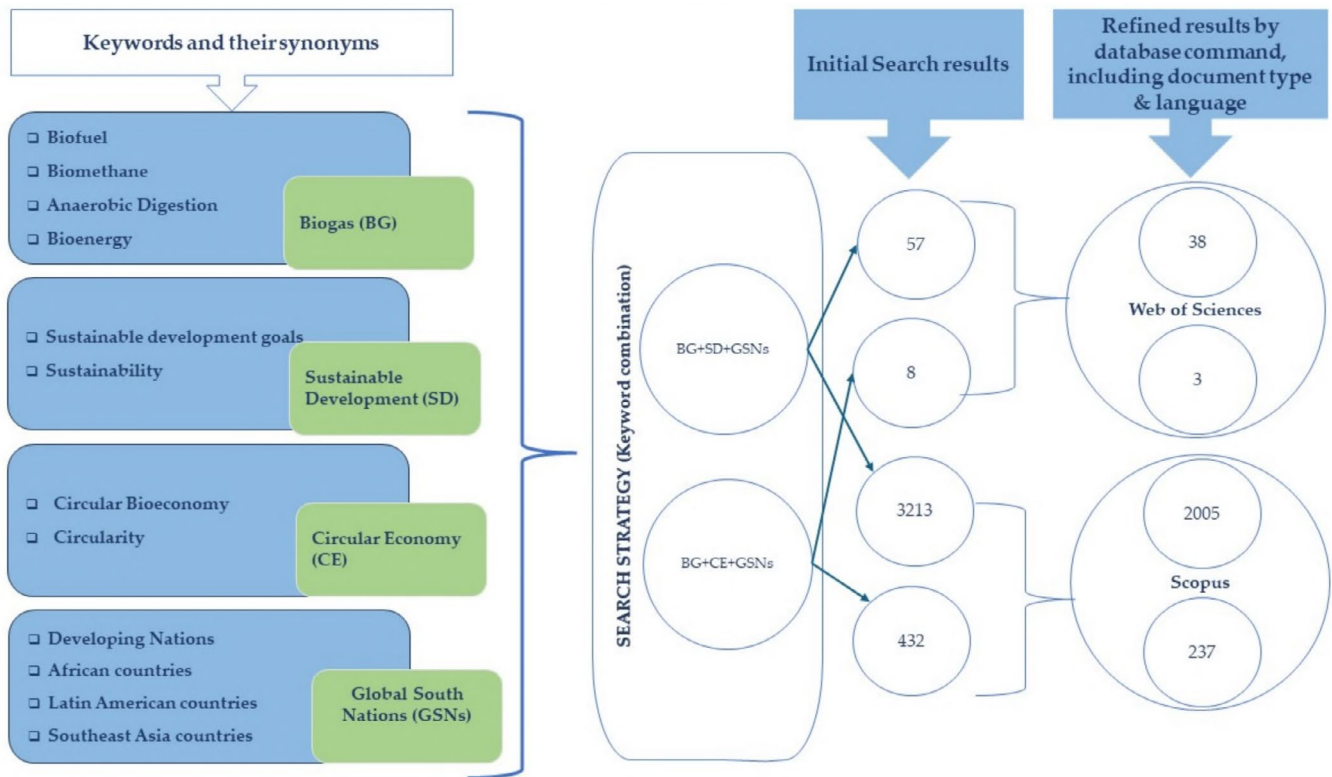


FIGURE 2 | Selected keywords, Search strategy (keyword combination for search), and their corresponding results. *Source:* Author.

1. Methodological approach.
2. Themes of the study.

As illustrated in Figure 3, a manual interpretative coding analysis framework has been developed by considering the aim of this study, the PRISMA 2020 protocol, current practical insights, alongside an extensive analysis of the existing literature related to the subject through repeated trials and enhancement. The framework facilitates the organization and categorization of data, thereby enabling the extraction of significant insights from complex information when discerning themes and patterns within the research examined. Predominantly grounded in contextual understandings to address the complexities of language that are crucial in the interpretative process (Isangula et al. 2024; Naeem et al. 2023; Neale 2016).

2.4 | Data Extraction

The methodical extraction of all relevant data concerning this research question from the incorporated peer-reviewed research articles was founded on the pre-existing manual interpretative coding framework, as illustrated in Figure 3. This process was executed iteratively, commencing with a deductive methodology that employs existing practical knowledge and a comprehensive review of the related literature, which then also serves to resolve differences when they arise. Consequently, standardized extraction forms/tables were developed, intended to aid the acquisition of supplementary

data as required, as presented in Tables 1 and 2. This is in alignment with the recommendation of Büchter et al. (2020), wherein the developed table was additionally applied to guarantee precision and uniformity in the coding of the requisite data, governed by the manual interpretative coding framework. The crafted extraction table underwent testing prior to commencing the actual data extraction procedure, providing extractors with practical experience with the form.

Consequently, Table 1 shows a sample of a comprehensive summary of the procedures employed for extracting data concerning methodological coverage. In alignment with this, the employed specific research designs are systematically categorized into three general research methodologies: quantitative, qualitative, and mixed methods (Strijker et al. 2020). Whereas Table 2 illustrates the thematic coverage of the research articles included. The data extraction process was facilitated using Mendeley Reference Manager and MS Excel to execute the review and compile the aggregated frequency data.

Collectively, in the processes of both methodological and thematic data extraction, the relevant information extracted included details concerning identified and assigned principal themes, major research methodologies, subthemes addressed, and employed specific research designs. Furthermore, it covered the frequency of articles categorized under each principal theme and methodologies. Additionally, the data encompassing article authors and titles are directly obtained from the original articles. This step facilitates the smooth progression to data analysis.

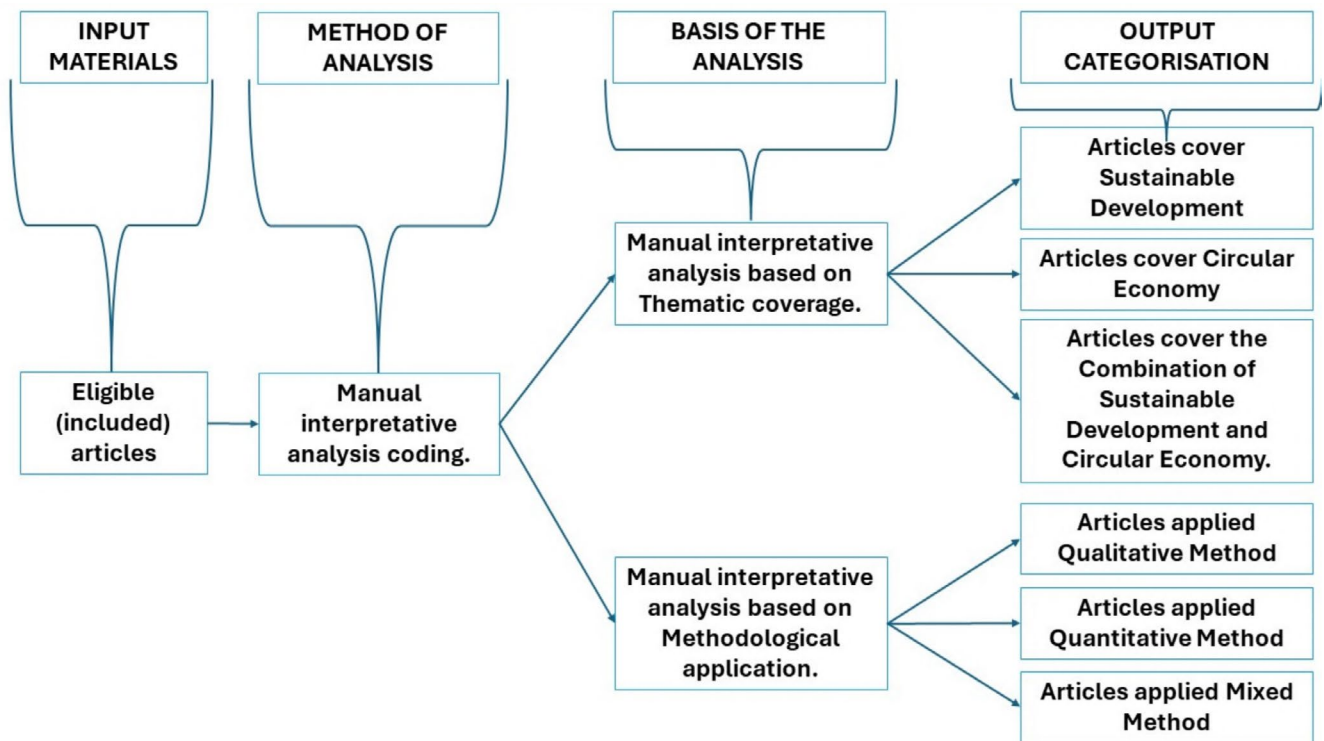


FIGURE 3 | A manual interpretative coding framework for thematic and methodological coverage analysis. *Source:* Author.

2.5 | Data Analysis

Likewise, the procedures for data extraction and data analysis were conducted based on a manual interpretative coding framework. This was achieved through the quantitative analysis of aggregated frequency data derived from the data extraction tables. To enhance understanding of the thematic alignment of the reviewed research articles with SD and CE, a bibliometric analysis was also performed using the biblioshiny interface in RStudio (Aria and Cuccurullo 2017). In the execution of the comprehensive methodological procedures, there are unequivocal strengths that merit preservation, along with specific limitations that should be addressed in future endeavors.

2.6 | Strengths and Limitations of the Study

The explicit strength of this study is its effort to delineate the bounds and expanse of the included peer-reviewed research articles by adhering to the PRISMA 2020 protocol and a standardized research reporting format, in maintaining procedural clarity, scientific rigor, and the quality of the review. In addition, it uses the benefit of expertise advantage by conducting a thorough feedback process between reviewers.

Even though the search strategy used two databases known for their capacity, based on a prior trial search, to cover extensive studies on the topic. This may limit the incorporation of studies from other databases and gray literature. Additionally, some peer-reviewed articles in these databases are inaccessible due to paywalls and language barriers, and preprints were excluded due to a lack of peer review. Despite careful keyword selection, some studies with different author keywords might be missed.

Furthermore, since this study followed a standardized research reporting format, it may not provide in-depth exposition on each theme that rests within its boundaries, like a thematically structured reporting format does. This is evident in the results and discussion chapters that follow.

3 | Results

The detailed investigation process, which adheres to the PRISMA 2020 protocol, facilitated the development of a search strategy through the combination of selected keywords for querying the two chosen databases: Web of Science and Scopus. This approach initially yielded 3710 results, as depicted in Figure 4. Following an initial refinement process utilizing the refining commands of the databases, the total was reduced to 2283 articles. Subsequently, as illustrated in Figure 4, the search results were further refined to 1907 articles by excluding duplicates and those marked as inaccessible. The refined articles underwent a title and abstract-level screening based on predetermined inclusion and exclusion criteria in a continuous iteration. This process resulted in the identification of 109 articles. At this point, numerous articles were identified as review articles, unrelated, and beyond the scope of this systematic scoping review.

A comprehensive screening of full articles was conducted on 100 articles retrievable from a pool of 109 qualified research publications. From this process, 41 peer-reviewed research articles were deemed suitable for inclusion in the study's analysis. Additionally, Figure 4 illustrates the entire procedure of refining the search results and the final output of articles incorporated. Then, according to the PRISMA 2020 diagram, a citation

TABLE 1 | A sample data extraction table for methodological coverage.

Roll no.	Article citation	Article title	Methodological content analysis and coding	Employed specific design	Major methodological category	Total # of articles
1	Armah et al. 2019	Biomethane potential of agricultural biomass with industrial wastewater for biogas production	Measuring the potential of biomass derived from agricultural byproducts for biogas production through the alteration of physicochemical properties.	Laboratory experiment	Quantitative method	29
2–18						
19	Laramée and Davis 2013	Economic and environmental impacts of domestic bio-digesters: Evidence from Arusha, Tanzania	Evaluating the impact of functional biogas digesters on the economic and environmental conditions of the local community.	Cross-sectional survey		
20–28						
29	Zhang and Qiu 2018	Comprehensive sustainability assessment of a biogas-linked agro-ecosystem: a case study in China	Evaluating the biogas-associated ecological subsystem within the framework of China.	Energy analysis by case study		
30	Bundhoo and Surroop 2019	Evaluation of the potential of bio-methane production from field-based crop residues in Africa	Evaluating the energy potential of crop residue for biogas production through theoretical frameworks and analysis of secondary data.	Assessment/theoretical estimation of biogas potential	Mixed method	11
31–39						
40	Nzila et al. 2012	Multi-criteria sustainability assessment of biogas production in Kenya	Evaluation of the Environmental, Technical, and Economic Aspects of Biogas.	A multi-criteria sustainability assessment		

(Continues)

TABLE 1 | (Continued)

Roll no.	Article citation	Article title	Methodological content analysis and coding	Employed specific design	Major methodological category	Total # of articles
41	Dhliwayo et al. 2020	Beyond climatic intervention: The social dimension of a biogas project in Sogwala village, Zimbabwe	A qualitative evaluation of the impact of the biogas project on the social dynamics within communities.	Qualitative study—social dimension	Qualitative/Descriptive method	3
42						
43	Tigabu et al. 2015	Technology innovation systems and technology diffusion: Adoption of bio-digestion in an emerging innovation system in Rwanda	An examination of the functional and historical progression of biogas technology as it pertains to its innovation and dissemination within Rwanda.	Longitudinal methodology/'process analysis'		

Source: Author.

reference was performed on 41 qualified articles to locate additional pivotal peer-reviewed research articles for inclusion in the analysis. The procedure initially identified 4 research articles, which were reduced to 2 after a thorough article screening, culminating in a total of 43 eligible peer-reviewed research articles for inclusion in this study. The detailed analysis is provided in the subsequent section, organized according to the predefined classification of the reviewed articles.

3.1 | Methodological Designs and Approaches

The manual interpretative coding analysis discloses the presence of diverse specific research designs within the examined peer-reviewed research articles. To facilitate comprehension, the outcomes are delineated at two distinct levels. Firstly, classification occurs at the level of three principal research methodologies, followed by the corresponding specific research designs. These specific research designs are aligned with the principal methods based on their standard association, thus enabling a quantitative analysis. Subsequently, Figure 5 illustrates the categorization of the 43 included peer-reviewed research articles across the three primary research methodologies according to each article's applied specific research design. Hence, the quantitative methods constitute the majority, accounting for 67%, followed by mixed methods at 26%, whereas qualitative/descriptive methods embody the smallest proportion at 7%.

The specific research designs employed in the evaluated peer-reviewed articles have been systematically classified into each principal research methodology (quantitative, qualitative, and mixed methods). The results are presented in Figure 6. Thus, as depicted in Figure 6. A., the subcategories of quantitative principal methodology, namely experimental and survey designs, represent the majority, comprising 13 and 12 peer-reviewed research articles, respectively. This includes articles by Armah et al. (2019); Chukwuma and Orakwe (2014); Djimtoingar et al. (2022); Ebeya et al. (2022); El Ibrahimy et al. (2022a); Getahun et al. (2014); Igbum et al. (2019); Kilucha et al. (2022); Kusmayadi et al. (2023); Ofoefule et al. (2011); Oyelaran et al. (2018); Sánchez et al. (2021); and Santos et al. (2022). Concerning the designated experimental designs, they seek to enhance anaerobic digestion by modifying physicochemical properties and biological components. The remaining four articles within this principal methodology pertain to a highlighted case study design.

Similarly, Figure 6B. unequivocally demonstrates 4 distinct specific research designs that are implied under the mixed method. Within this category, survey design emerges as the most prevalent approach, being employed in six peer-reviewed research articles. This includes the studies conducted by Bundhoo and Surroop (2019), Chen et al. (2023), Kefalew and Lami (2021), Mboumboue and Njomo (2018), Shaibur et al. (2021), and Singh and Kalamdhad (2022). Then, a case study design was followed, applied in two articles. The distinction between the survey and case study design under this principal methodology, and the one stated under quantitative research, lies in the integrated application of numerical analysis alongside the narrative exposition of non-numerical data. The remaining specific designs categorized under mixed methods are environmental impact assessment

TABLE 2 | A sample data extraction table for thematic coverage.

Roll no.	Article citation	Article title	Content analysis and coding	Subthemes addressed in the articles	Major theme	Total # of articles
1	Ahmad and Jabeen 2023	Biogas technology adoption and household welfare perspectives for sustainable development	Evaluating the relationship between household welfare and sustainable development consequent to the adoption of biogas technology.	◦ Economic dimension, ◦ Social dimensions, and ◦ Environmental dimension.	Sustainable Development (SD)	17
2–16						
17	Wang et al. 2020	Determining the influencing factors of biogas technology adoption intention in Pakistan: The moderating role of social media	Evaluating the impact of social media on the decision-making processes of local communities regarding the adoption of biogas technology.	◦ Social dimensions		
18	Armah et al. 2019	Biomethane potential of agricultural biomass with industrial wastewater for biogas production	Assessing agricultural biomass potential for biogas production by altering physicochemical properties.	◦ Waste-to-energy ◦ Waste management	Circular Economy (CE)	10
19–26						
27	Santos et al. 2022	A biochemical methane potential of pig slurry	Assessing biogas production potential from pig slurry using two mixture formulations.	◦ Waste-to-Energy		
28	Bundhoo and Surroop 2019	Evaluation of the potential of bio-methane production from field-based crop residues in Africa	Evaluation of the energy potential of crop residues for biogas production, predicated on theoretical models and secondary data.	◦ Environmental dimensions ◦ Waste-to-Energy.	The combination of SD and CE	16
29–42						
43	Zhang and Qiu 2018	Comprehensive sustainability assessment of a biogas-linked agro-ecosystem: a case study in China	Evaluation of the biogas-associated ecosystem subsystem within the context of China.	◦ Waste management ◦ Environmental dimension ◦ Economic dimension		

Source: Author.

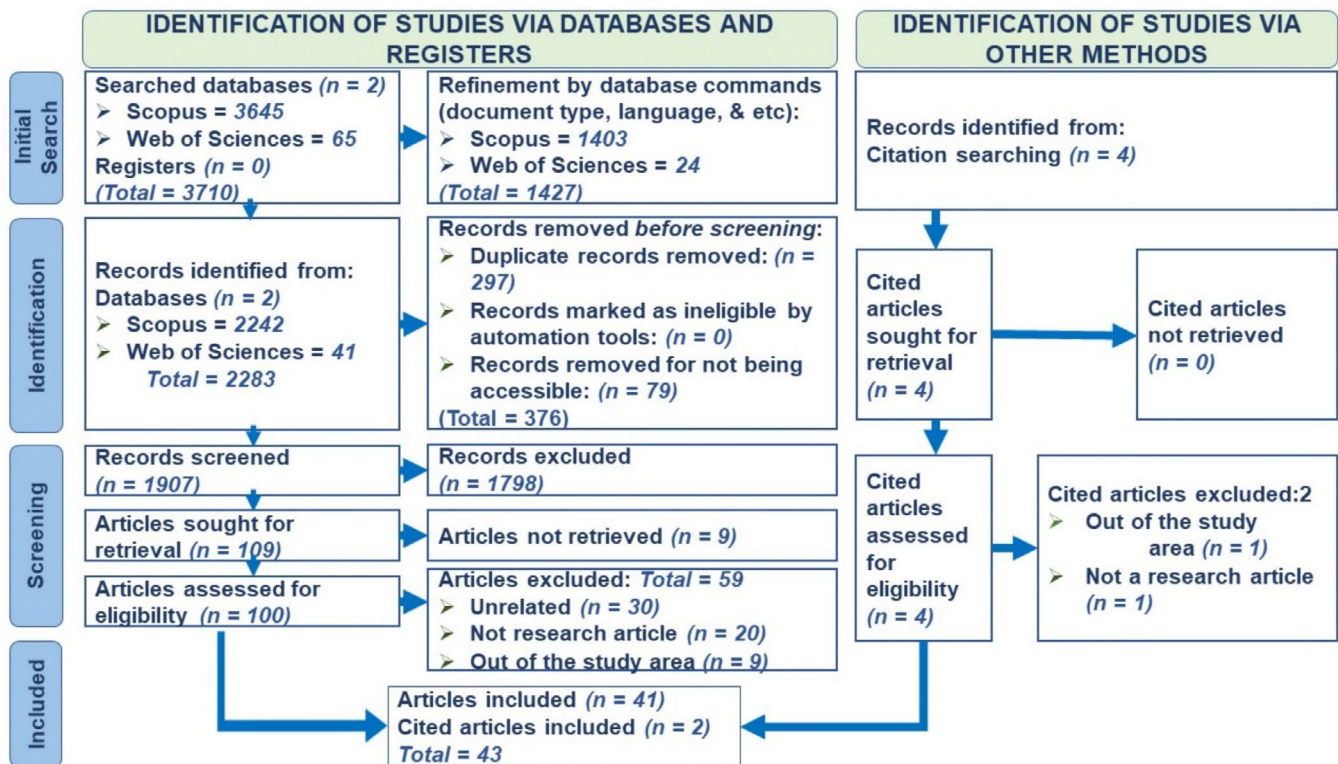


FIGURE 4 | PRISMA 2020 flow diagram search result. *Source:* Adopted and modified from Page et al. (2021).

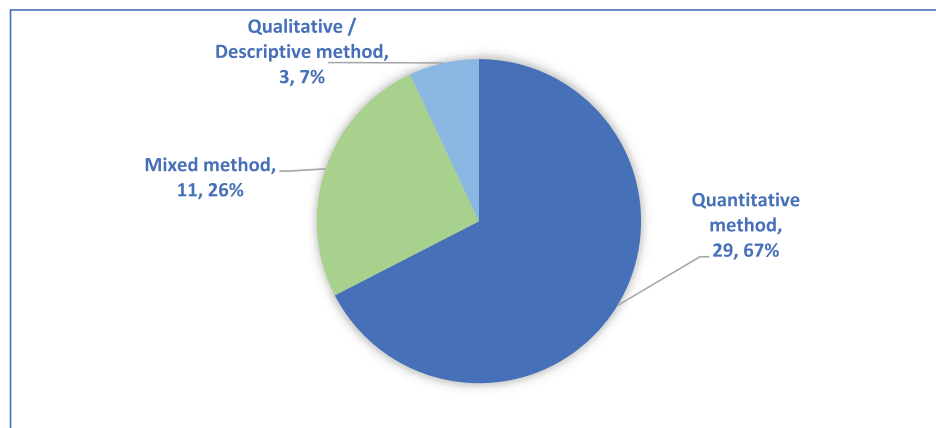


FIGURE 5 | A manual interpretative coding analysis classification of principal methodologies and their coverage. *Source:* Author.

and a life cycle assessment study. Each of them is represented by 1 article.

Ultimately, prior to engaging in the thematic analysis, Figure 6. C. elucidates specific research designs inherent within the qualitative methodology. This methodology comprises two distinct research designs: project evaluation design and process analysis design. The project evaluation design includes 2 peer-reviewed articles authored by Dhliwayo et al. (2020) and Tembo et al. (2022). Whereas the process analysis design encompasses 1 peer-reviewed article by Tigabu et al. (2015). Collectively, these designs concentrate on biogas project performance evaluation and the investigation of historical and functional advancement of biogas technology.

3.2 | Themes of Studies

The thematic analysis uses dual approaches to clearly understand the core thematic coverage of the investigated peer-reviewed articles. The primary method utilized is a crafted table informed by a manual interpretative coding framework. To do that, three thematic categories were used to classify the research articles based on their alignment. In addition, the second approach for scrutinizing thematic coverage involves applying bibliometric analysis, utilizing the authors' designated keywords. In contrast to the manual interpretative coding analysis, bibliometric analysis is executed using RStudio, particularly via the biblioshiny interface. The findings are delineated in the subsequent subsections.

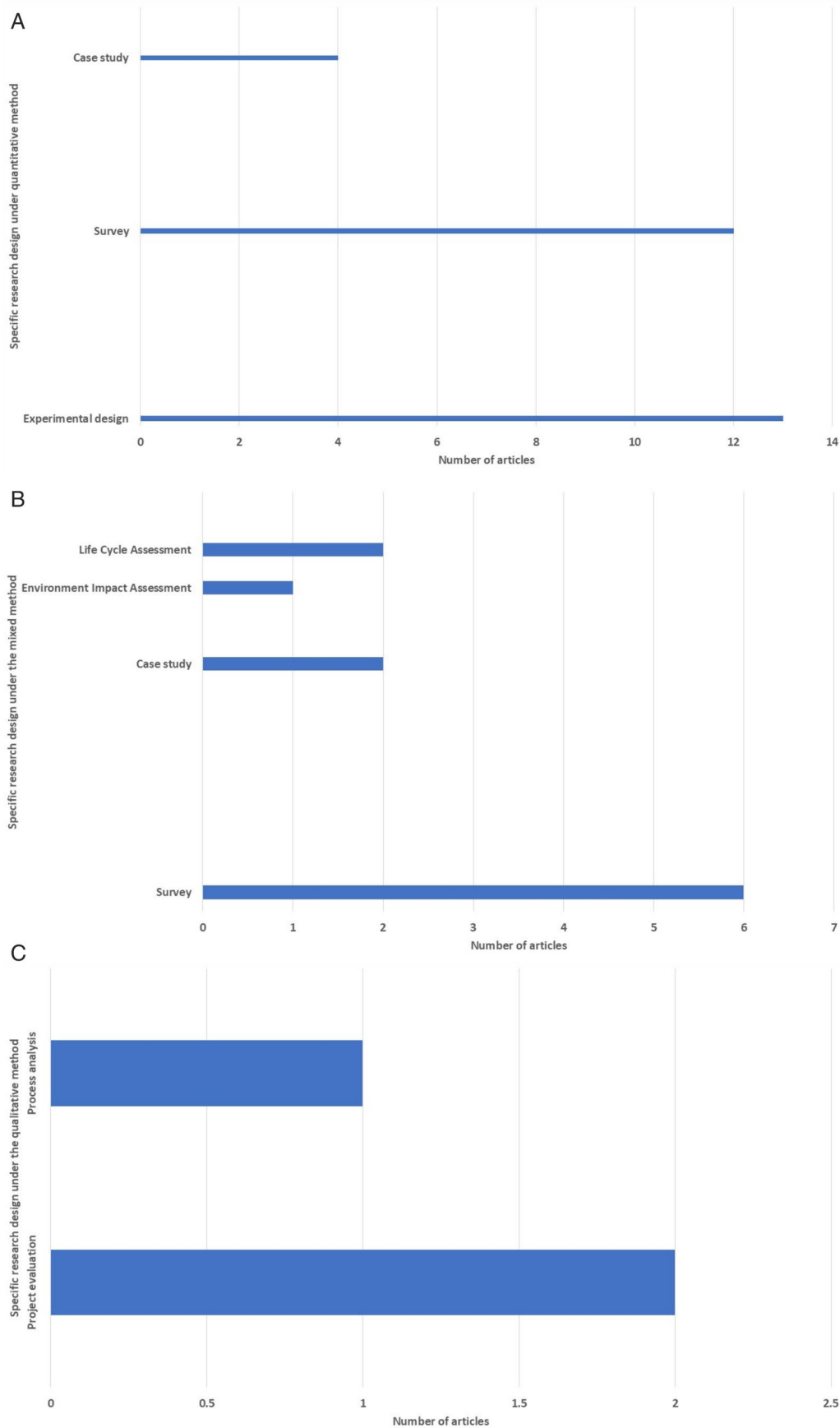


FIGURE 6 | A distribution of each specific research design under the principal methodology. (A) Distribution of specific research designs under the quantitative research method. (B) Distribution of specific research designs under the mixed research method. (C) Distribution of specific research design under the qualitative research method. *Source:* Author.

3.2.1 | A Manual Interpretative Theme Analysis

Figure 7 presents the three predefined thematic categories (SD, CE, and the intersection of SD and CE) along with their respective coverage proportions within the reviewed peer-reviewed articles. Therefore, as illustrated in Figure 7, 40% of the reviewed articles integrate SD and the CE as a distinct theme. Following this, the SD theme constitutes 37%. Although it is uncommon to establish a direct correlation, the remaining 23% are associated with the theme of the CE.

Likewise, in the methodological analysis, where the specific research designs identified from the investigated peer-reviewed articles inform the principal methodologies, subthemes are also used to infer the principal themes. This is demonstrated in Figure 8. Thus, the principal theme, which integrates SD and the CE as a unified theme within a single scholarly article, encompasses a total of six subthemes. Consequently, as revealed in Figure 8A., the environmental subtheme receives the most extensive coverage, appearing in 15 peer-reviewed articles, followed by the waste-to-energy subtheme, which is covered in 12 peer-reviewed articles. Continuing in this rank, the waste management subtheme occupies the third position with coverage in 9 peer-reviewed articles. Subsequently, the economic subtheme ranked fourth, being addressed in 5 peer-reviewed articles. Finally, both the social dimension subtheme and the technical/technological dimension subtheme were covered in 2 peer-reviewed articles.

Conversely, the principal theme of SD encompasses four distinct subthemes, as illustrated in Figure 8B. Within this primary theme, the economic dimension subtheme receives the most extensive coverage, being addressed in 12 peer-reviewed

articles. In contrast, the social and environmental dimension subthemes are each found in 7 peer-reviewed articles. The technical/technological dimension subtheme demonstrates the least prevalence, being covered in 3 peer-reviewed articles. Furthermore, concerning the principal theme of the CE, two subthemes are established beneath it. It is found to be the theme with the fewest subthemes. As revealed in Figure 8C, the waste-to-energy subtheme is notably prominent, as it is addressed in 10 peer-reviewed articles. Whereas the waste management subtheme ranks second, being discussed in 3 peer-reviewed articles.

In addition, it is pertinent to note the quantitative difference between the aggregate number for each primary theme and the cumulative number of peer-reviewed articles for their corresponding subtheme. Appendix A at the end of this paper elucidates this and facilitates a deeper comprehension of the matter. For instance, as illustrated in the same Appendix A, roll number 1 under the principal theme of SD and CE encompasses 17 peer-reviewed articles; meanwhile, the cumulative total of its corresponding subthemes amounts to 47 peer-reviewed articles. This discrepancy in the aggregate figures is attributed to how the review process is conducted and the extent to which the peer-reviewed articles address each subtheme. Hence, the practical interpretation of this circumstance is the incorporation of multiple subthemes in a single peer-reviewed article.

3.2.2 | A Bibliometric Theme Analysis

The outcomes of the bibliometric analysis are presented in Figure 9. Though Figure 9A. reveals that three keywords—biogas, anaerobic digestion, and renewable energy—exhibit a

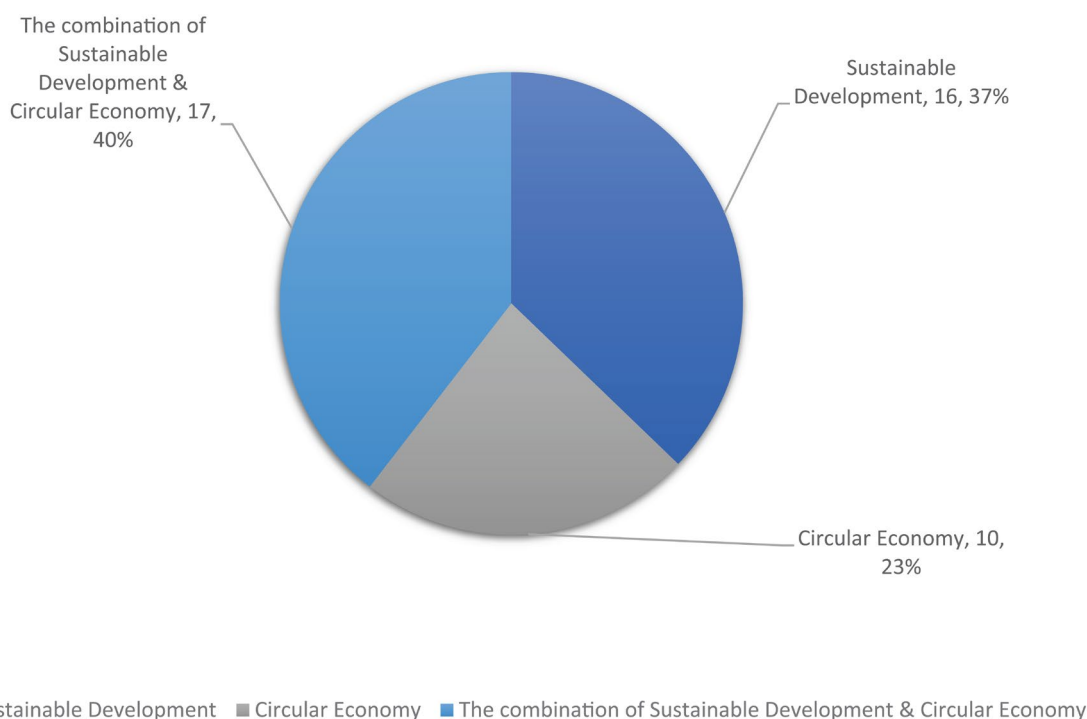


FIGURE 7 | A manual interpretative coding analysis classification of principal themes and their coverage. *Source:* Author.

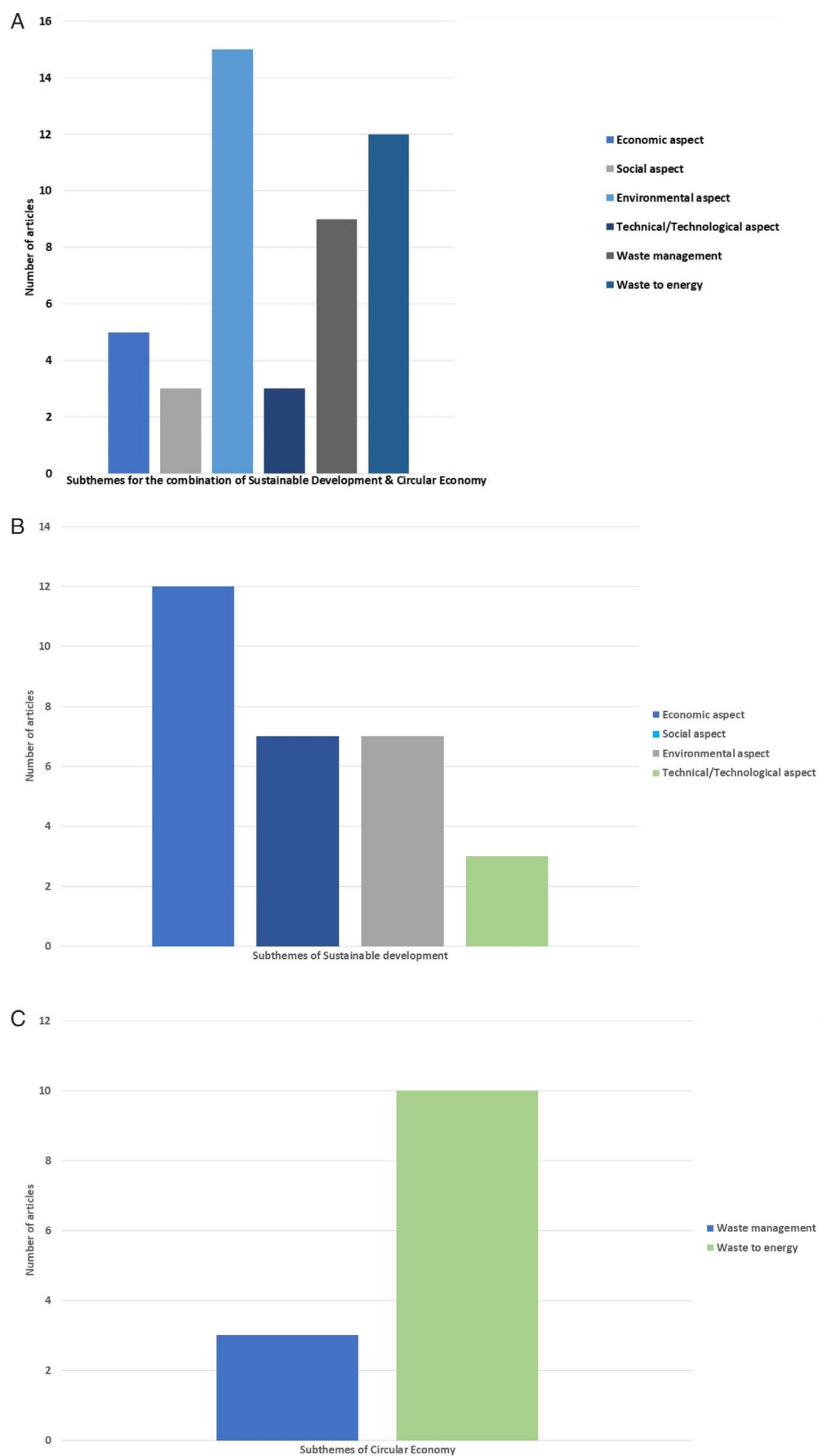


FIGURE 8 | A distribution of each subtheme under the principal themes. (A) Distribution of subthemes under the principal theme that combines SD and CE. (B) Distribution of subthemes under SD. (C) Distribution of subthemes under CE. *Source:* Author.

significant development as themes in terms of both relevance and degree of development over time. Conversely, the themes that have experienced a decline in both relevance and degree of development over time include municipal solid waste, biodegradability, and livestock manure. Nonetheless, the primary focus of this systematic scoping review, namely SD and the CE, failed to emerge as either a developing or diminishing theme in both relevance and degree of development over time.

Alternatively, Figure 9B. demonstrated that the terms “biogas,” “anaerobic digestion,” and “renewable energy” exhibit the most prominent appearance due to their frequent occurrences as author-designated keywords. The extensive associations and connection with various terms/keywords further illustrate their recurrent co-occurrences. However, the presence of SD and CE or terms related to them, along with associated themes, remains notably sparse, even with a limited number of recurrences and linkages.

The results pertaining to thematic coverage, as derived from the methodologies applied, pave the way to elucidate the third crucial aspect of this research, which addresses the regional circumstances of the Global South, as detailed in the following section.

3.3 | Bibliometrics Analysis of the Global South Nations' Context

3.3.1 | Global South Nations' Contribution to Scientific Studies

The scientific engagement of Global South Nations on biogas technology, aiming to advance SD and CE, is investigated through a bibliometric analysis. The outcomes of this analysis are illustrated in Figure 10.

Thus, Figure 10 illustrates the contribution of 31 countries as lead for the publication of 43 peer-reviewed research articles. Notably, the number of leading nations in the publications seemed to be fewer than the number of articles published. However, the total number of countries contributing at different levels is 60. On the other hand, among the 31 leading nations, 21 countries are from the Global South. The remaining 10 are Global North Nations. Accordingly, the top five Global South Nations from the lead countries include China, with 31 peer-reviewed articles, occupying the first position. South Africa, followed by 8 peer-reviewed articles. Nigeria holds the third position with 6 peer-reviewed articles. Then, Ethiopia, Mauritania, and Pakistan share the fourth position, each with 5 peer-reviewed articles. On the contrary, at the bottom of this ranking, eight countries are represented with a single peer-reviewed research article. These include Botswana, Colombia, Nepal, Tanzania, Zambia, and Zimbabwe.

Moreover, Figure 11 illustrates the Global South Nations' scientific output spreads on a global scale, along with the existing cooperation within the Global South and beyond.

A citation analysis result among the 31 leading publishing countries is presented in Figure 11A. It revealed that South Africa, the second in publishing, stood first with 101 citations. Pakistan,

Cameroon, and Ethiopia are the second, third, and fourth with 51, 45, and 38 citations, respectively. Even if China was the first in publishing, it ranked fifth with 32 citations. In the case of international collaboration in scientific publication, as depicted in Figure 11B. China is at the forefront with five outputs. South Africa and Pakistan followed in the second place with two outputs each. Bangladesh and Ethiopia are positioned at the lower end of the list, each contributing one peer-reviewed article. Notably, Pakistan's contribution comprises solely a multi-country publication. Whereas Cameroon, Morocco, and Nigeria produced a single-country contribution publication of 2 peer-reviewed articles, respectively.

3.3.2 | Global South Nations' Publication Throughout the Years

This systematic scoping review incorporates trajectories of publication over the years, extending from 1988 to April 2023. It is because of the introduction of the term ‘sustainable development’ into global development political discourse in 1987. Nevertheless, the bibliometric analysis results in Figure 12B. indicated that the initiation of scientific inquiry was marked in 2007, later than initially projected. Then it shows an overall increasing trend in the volume of publications. The notable exceptions were in the years 2015–2016 and 2020.

In contrast, Figure 12A. presents the top 6 countries based on the ranking of individual countries' publications over time. South Africa embarked on this initiative early, consistently maintaining its position with a steady and modestly inclined trend. Conversely, China remained relatively restrained until around 2017. Subsequently, following 2017 for the five consecutive years, China's publication trend aligned with a standard upward trajectory like other countries listed. This pattern underwent a significant transformation in 2022, characterized by a noteworthy, sharply accelerating increase. The other four countries, namely Nigeria, Ethiopia, Pakistan, and Mauritania, commenced their scientific publishing sequentially and are observed to exhibit analogous moderate upward trends.

3.3.3 | Global South Nations' Relevant Authors and Their Affiliation

Alongside clarifying the trend in publications and the scientific contributions of diverse Global South countries over the years, the bibliometric analysis also illustrates the relevant authors and associated institutions based on the number of publications. This analysis delineates the nexus among the pertinent authors, affiliated institutions, and their countries. Figure 13 portrays these facts.

Moreover, Figure 13 reveals that only 10 authors were recognized as the most prominent, based on their publication count. Each author has contributed 2 peer-reviewed research articles. Consequently, this reflects the publication of 20 peer-reviewed research articles by these 10 authors, indicating that approximately 47% of the entire publication output is attributed to this handful of authors. In conclusion, a comprehensive understanding of the study's findings establishes a foundation for the

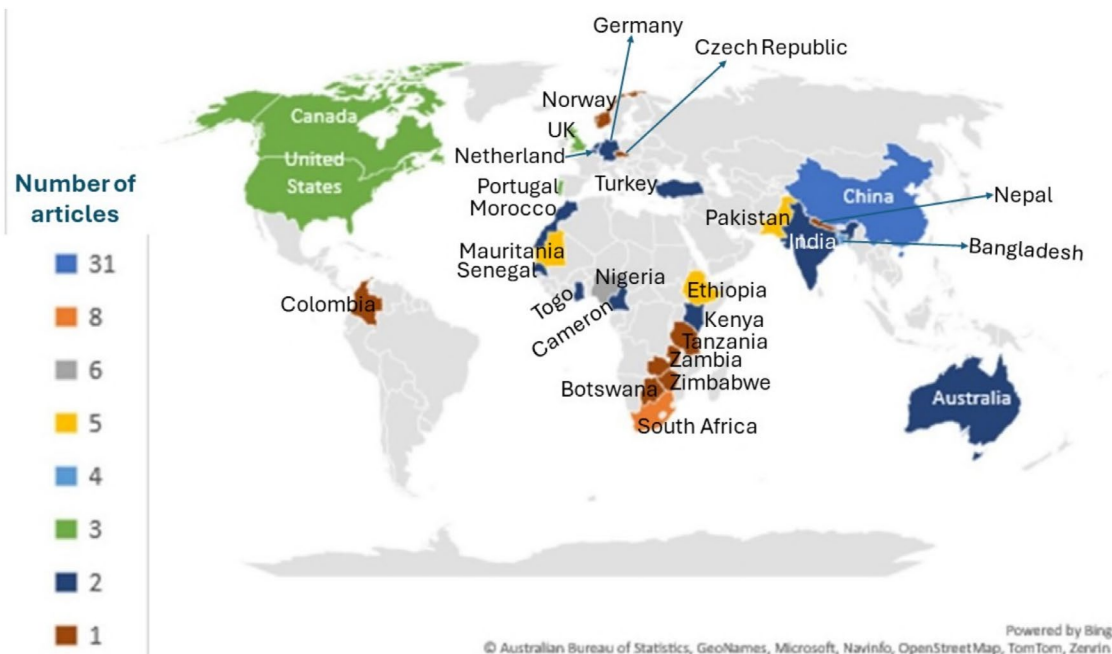


FIGURE 10 | Identified leading countries and their contribution to scientific publications. *Source:* Author.

subsequent discussion chapter, which will contextualize our results within the framework of existing literature.

4 | Discussion

In this chapter, each sub-section is devoted to restating and interpreting the findings, accompanied by a succinct explanation that aligns with the research question. This procedure entails comparing with preexisting studies, emphasizing both commonalities and disparities, to explicate their ramifications for policy and practice.

4.1 | Research Methodological Trends

The results of this study convey that researchers adopt various research methodologies and designs in their investigations of biogas within the context of SD and CE. Nonetheless, as revealed in the findings, the quantitative approach represents the predominant methodology. This assertion is supported by the findings of Horvath et al. (2022), which highlight the widespread employment of quantitative methods in scientific research concerning SD. Moreover, within the quantitative method, experimental design is predominant in examining advanced anaerobic digestion options through the manipulation of physicochemical characteristics and biological components (Kilucha et al. 2022; Kusmayadi et al. 2023; Santos et al. 2022).

However, Taherdoost (2022) stated that although quantitative approaches are crucial, relying too heavily on them can result in a narrow perspective regarding the challenges and opportunities in the discipline. It is essential to acknowledge the limitations and potential biases of this approach. It might also overlook the valuable insights provided by qualitative research. For example, qualitative methods such as case studies

and ethnography can uncover the complexities of human behavior, societal impacts, and contextual elements that quantitative data cannot adequately address (Caggiano and Weber 2023; Keith et al. 2022; Taherdoost 2022). Furthermore, the experimental designs predominantly utilized in quantitative research frequently emphasize controlled variables. This emphasis may not accurately represent real-world circumstances, where multiple factors interact in unforeseen ways (Leatherdale 2019). Such an approach could result in a limited understanding of biogas systems and their role within the CE, consequently jeopardizing the thorough evaluation required for SD (Kapoor et al. 2020; Leatherdale 2019; Suárez-Eiroa et al. 2019; Taherdoost 2022).

Even though quantitative methods continued to be employed extensively across a wide array of disciplines because of the diverse design options they offer, setting them apart from qualitative and mixed methodologies (Queirós et al. 2017). Mulisa (2022) highlights that specific critical scenarios necessitate the application of quantitative methods. Such scenarios encompass the characteristics of the data, the aim to achieve high external validity (generalizability), the study's objective stance towards confirmatory inquiries with substantial data, the replicability of these methods, and the availability of resources. However, Jamieson et al. (2023) propose that researchers utilizing quantitative methods should practice reflexivity by critically assessing their presuppositions throughout all stages of research. While Queirós et al. (2017) observed, in agreement with our findings, that qualitative methods are used less frequently in general, they found them to be particularly effective in the context of social care studies. Citation analysis shows the significant impact of the method used. The selection of these methods is contingent upon certain specific conditions. According to Mulisa (2022), these conditions involve the demand for strong internal validity; an emphasis on exploratory subjectivity for understanding context, meaning,



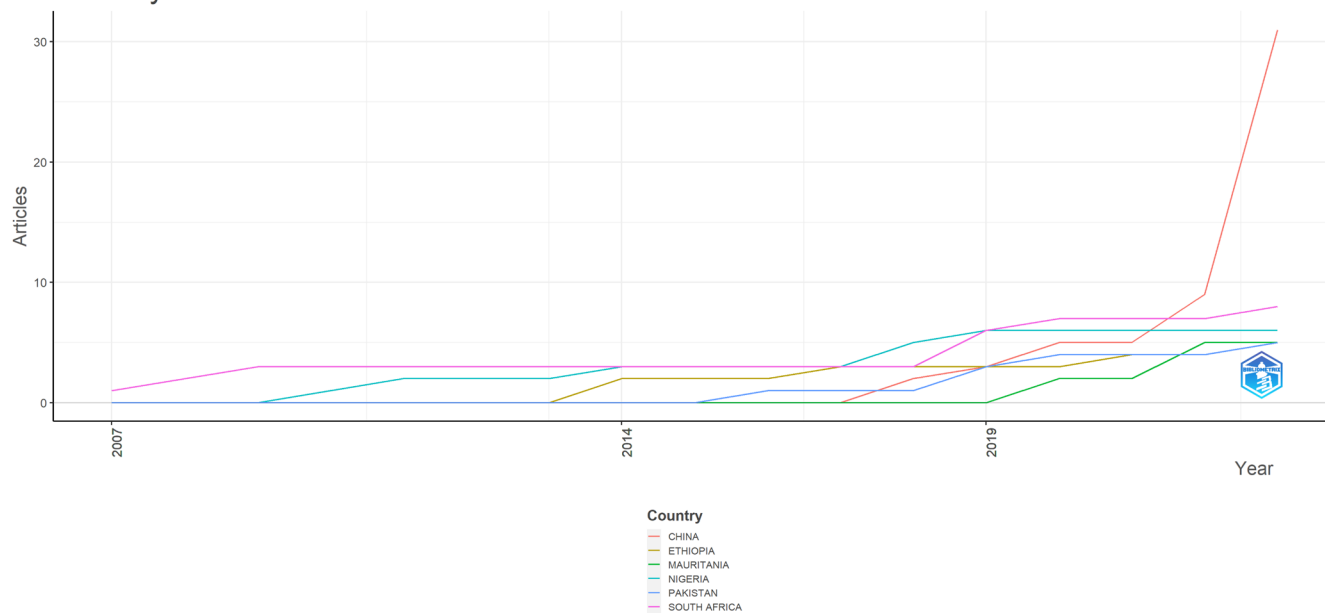
FIGURE 11 | Cooperation between Nations and the highest total citations for countries' publications. (A) The highest total citations for countries' publications, (B) Cooperation between Nations on publications. *Source:* Author.

and process; the need to operate within real-world environments; capturing the perspectives of participants; and using a smaller sample size.

Mixed and qualitative methods are crucial for developing policies and partnerships in biogas technology, contributing to sustainable development by implementing circular economy models (Chen et al. 2023; Singh and Kalamdhad 2022;

Tembo et al. 2022). Combining qualitative and quantitative methods enhances understanding and provides a detailed view for comprehending the intricate interactions and cross-disciplinary subjects (Kinnebrew et al. 2021; Molina-Azorín and López-Gamero 2016). This approach assesses social acceptance and behavioral factors in biogas adoption, aligning with the principles of a circular economy, such as efficient resource use (Taherdoost 2022; Tembo et al. 2022). Nonetheless, it is

A Country Production over Time



B Annual Scientific Production

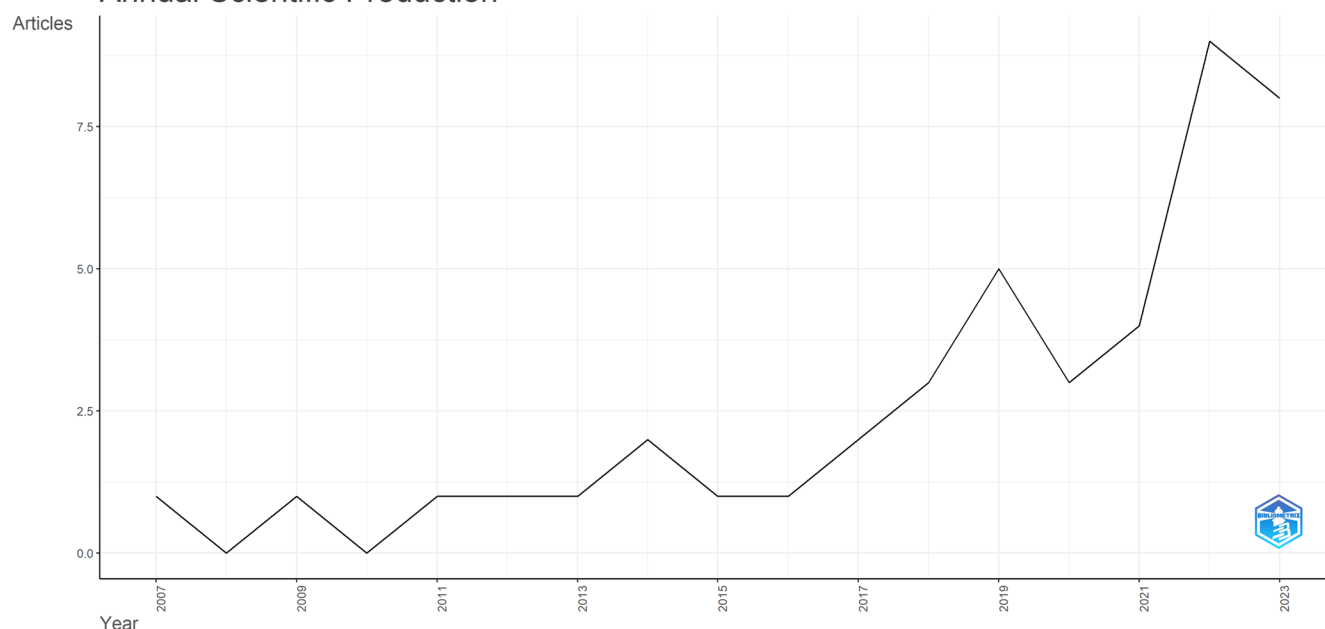


FIGURE 12 | Publication trend of the top six publishing Global South Nations and the overall publication trend throughout the study period. (A) Publication trend of the top six publishing Global South Nations, (B) the overall publication trend throughout the study period. *Source:* Author.

not necessarily optimal for studies predicated on an alternative methodology (Molina-Azorín and López-Gamero 2016). Empirical case studies show that public opinion is vital for renewable energy projects, especially with community participation (Dhliwayo et al. 2020). Furthermore, as Wang et al. (2020) indicated the integration of rapidly evolving and advancing information technology (IT) and Internet of Things (IoT) products into the design and application of research methodologies has been determined to be limited. Therefore, Mishra et al. (2022) and Motlagh et al. (2020) endorse the application of the Internet of Things, digitization, and assorted applications that advance methodological practices and enhance the production and

consumption of renewable and green energy, consequently fostering sustainable development.

4.2 | Identified Thematic Trends

The shift from traditional to renewable and sustainable energy sources, including biogas technology, constitutes a central theme in contemporary scientific discourse (Kefalew and Lami 2021; Kilucha et al. 2022). Nonetheless, the results of this study reveal that reviewed scientific studies are frequently undertaken without a comprehensive integration of CE principles or the SD

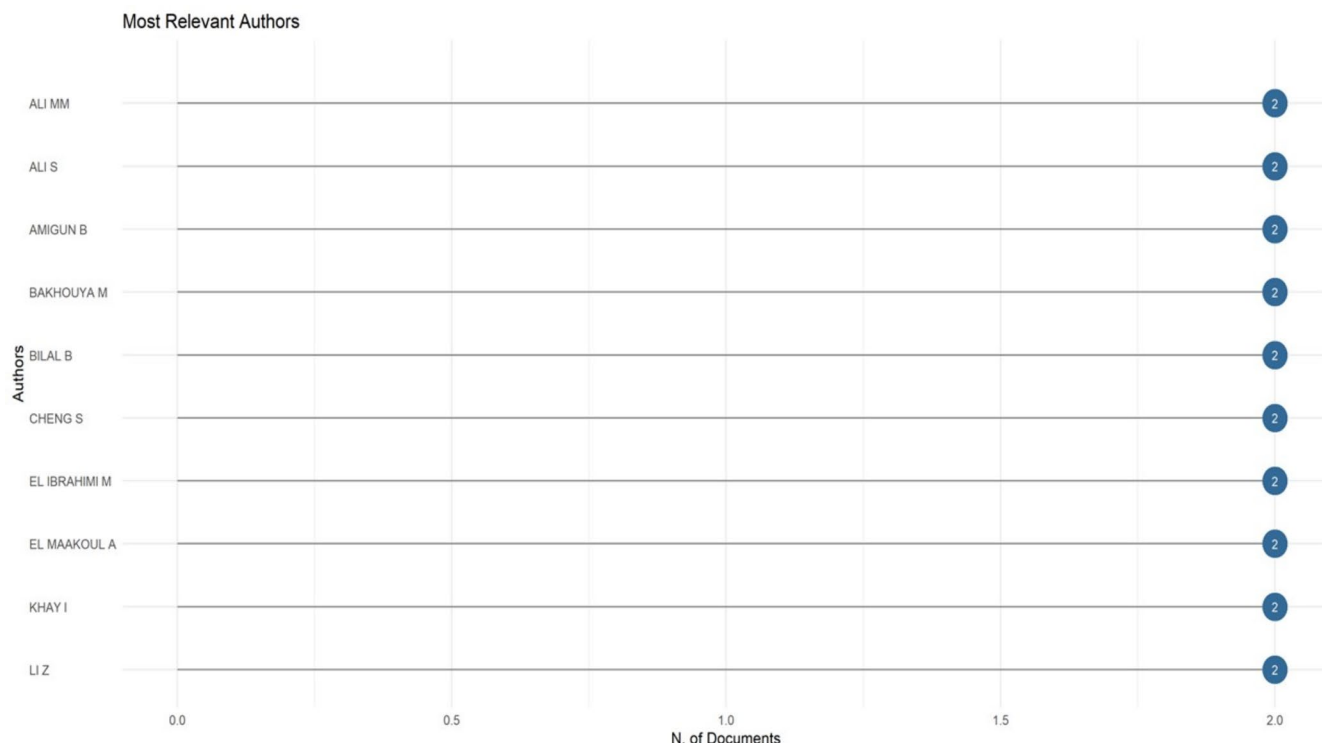


FIGURE 13 | Top publishing authors (Most relevant author by number of publications). *Source:* Author.

framework, which encompasses economic, social, and environmental dimensions. As Ali et al. (2023) noted, this is the result of focusing on short-term benefits to tackle urgent environmental issues, stemming from past and present energy use, which could worsen environmental degradation and social inequality, threatening future generations' prosperity. However, D'Aquino et al. (2022) have suggested that transitioning to sustainable energy solutions that promote resource circularity to minimize waste, including biogas technology, is vital not only for mitigating these negative outcomes but also for enhancing economic resilience and elevating the quality of life for affected communities. Consequently, in the following sub-section, the findings of this study on the identified themes are presented through the lens of the extant literature.

4.2.1 | Trends of SD as a Theme

This review indicates that in the Global South, SD is not prominently featured as a central theme in the biogas studies under consideration, despite its significance. Theoretically, as Güney (2021) posits, this should not be the scenario since SD markedly prioritizes and supports renewable energy over non-renewable energy sources, including biogas. The research frequently explores the benefits of biogas, such as economic, social, and environmental impacts (Ali et al. 2020; Bakehe 2021), yet does not establish a cohesive SD theme. Similarly, as evidenced in our findings, the economic, social, environmental, and technological dimensions of SD are examined independently by using their futures in an unstructured manner. This is insufficient for recognition as a theme in the comprehensive context of SD. However, economic and social dimensions featuring the benefits of biogas appear equally in the reviewed articles, but they do so indirectly.

Hence, our study like the findings of (Obaideen et al. 2022; Yasar et al. 2017) illustrates that biogas' economic dimension for SD futured now and then by job creation, cost-effective energy production for rural areas, improving energy security, and reducing fossil fuel reliance, but do not explicitly fit it within SD frameworks. In the same way, as disclosed by Verner et al. (2023) and Yasar et al. (2017), the social dimension, associated with the enhanced quality of life, health, and community empowerment. Beyond, in agreement with the findings of Obaideen et al. (2022) and Yasar et al. (2017), it also serves as a mitigator of indoor pollution, eases the burden of firewood collection on women and children, and promotes education and income generation, fostering gender equality and community involvement in sustainable practices. However, in Global South Nations, challenges like high costs, lack of awareness, and ineffective business models hinder these benefits (Archana et al. 2024). For instance, Nevzorova and Kutcherov (2019) outline six challenge domains, including technical, economic, market-related, socio-cultural, institutional, and environmental. Each includes barriers that affect the economic and social dimensions. To illustrate, Mittal et al. (2018); Tembo et al. (2022) listed barriers like high investment costs, poverty, lack of subsidies, competition with alternative energy, insufficient research & development funding, low fossil fuel prices, political instability, lack of public-private sector coordination, limited public engagement, stigmatization, resistance to change, and migration issues.

Similarly, this study did not identify any direct association between the environmental dimension of SD and biogas within the reviewed research articles. Instead, as stated by Nethengwe et al. (2018) and Uhunamure and Shale (2021), biogas, described as a form of renewable energy, serves as a low-carbon

alternative to fossil fuels. This insight led to the identification of renewable energy as a significant theme in our findings through bibliometric analysis. The term renewable energy represents one of the environmental benefits of biogas, functions as a dependable, low-emission energy source, and directly promotes SD (Atchike et al. 2022; Nwankwo et al. 2024). Nevertheless, Bishoge et al. (2020) observed that, in practical terms, numerous environmental obstacles are obstructing Global South Nations from fully exploiting their abundant renewable energy resources. Particularly, barriers like odor complaints and demand for substantial water resources are prevalent (Nevzorova and Kutcherov 2019). Nevertheless, if these obstructive challenges are surmounted, biogas can reduce carbon emissions significantly (Subbarao et al. 2023). Consistently, its effectiveness in waste management emerges as a significant environmental benefit, as it employs organic waste (e.g., agricultural residues, food waste, animal manure) in biogas production, thereby reducing reliance on landfills and greenhouse gas emissions (Mittal et al. 2018).

In this investigation, the technical/technological dimensions of SD are the least prominent among the four dimensions. The technical/technological dimensions relate to scientific and technological advancements that foster economic growth while preserving environmental integrity (Ahmad et al. 2023). These attributes emerged from studies focusing on physicochemical and biological manipulation to enhance anaerobic digestion and biogas production. This includes the pre-treatment of feedstock using locally accessible digestion-accelerating biomass, which is advocated to enhance the chemical, physical, thermochemical, and oxidative yields of biomethane and biogas (Sher et al. 2024). Concerning system design, the development and application of prefabricated biogas digesters are increasingly being explored, assessed, and adopted in developing countries as a means to overcome the limitations of traditional domestic digesters, including the fixed dome, floating drum, and plug flow models. prefabricated biogas digesters are typically classified into two types: composite material digesters (CMDs) and bag digesters (Cheng et al. 2014).

However, the implementation of biogas technology, notwithstanding its numerous advantages and benefits, is not without certain limitations that warrant careful consideration. This includes the necessity for maintaining a consistent temperature that presents substantial challenges when implementing the technology in regions with significant seasonal temperature fluctuations (Garkoti et al. 2024). This predicament is particularly evident in rural China, where temperature variability detrimentally impacts the efficiency of digesters and biogas production (Li, Wan, et al. 2023). The challenge could be mitigated through the deployment of heating systems that exploit biogas, solar, and geothermal energy, selected based on their economic feasibility and cost-effectiveness (Garkoti et al. 2024). Technical barriers like infrastructure issues, poor waste management, scarcity of local biogas tech, lack of trained personnel, and frequent repairs are significant challenges for Global South Nations (Ali et al. 2022; Farrukh et al. 2024). In debunking this, future smart metering and remote monitoring systems are also expected to transform biogas plants, including the small-scale ones (Ntaganda et al. 2024).

Therefore, over the past decade, within the framework of SD, as the global paradigm transitioned from the MDGs to the SDGs (United Nations—General Assembly 2000; United Nations 2015) the global community has been earnestly striving to adopt clean, sustainable, affordable, and renewable energy sources that advance SD and the realization of the SDGs (Ali et al. 2023; Jin and Zafar 2024; Bundhoo and Surroop 2019). This asserts that biogas' current prominent position in research is due to its potential to fulfill renewable energy requirements (Atchike et al. 2022; Uhunamure and Shale 2021). Furthermore, despite a non-direct association of biogas with SD dimensions in this investigation, on the other hand, anaerobic digestion is also highlighted as a pivotal theme in our bibliometric analysis findings.

4.2.1.1 | Trends of SDGs in Biogas Studies. The principal objective of the study persisted as the evaluation of the research methodologies employed, alongside an analysis of thematic trends, particularly SD and CE, within the articles reviewed. This focus is due to the sporadic presence and conspicuous absence of SDGs at a level of thematic prominence. However, omitting a discourse on the critical global policy framework of SDGs, which actively facilitates the implementation of CE and the attainment of SD, would be considered inappropriate. Consequently, here we provide a thorough exposition of the SDGs with the thematic findings of this study.

Initially, to elucidate the ambiguities surrounding the SDGs and to achieve an equitable comprehension thereof, it is pertinent to reference the assertions of Fehling et al. (2013) and Sachs (2012) regarding the SDGs. Despite the comprehensive processes undertaken for their formulation and implementation as a pivotal global policy framework, the SDGs have encountered a myriad of criticisms from the international community. Yet simultaneously serve to unify developmental discourse. Critics hold a spectrum of views, predominantly concentrating on the tension between the SDGs' 3% annual economic growth target and environmental sustainability (Gabay and Ilcan 2017; Hickel and Jason Hickel 2019). Nevertheless, McCollum et al. (2018) contend that the positive interactions of the SDGs surpass the negatives. In all scenarios, however, it remains essential to emphasize varied worldviews beyond 'modernity,' as such an emphasis promotes grassroots participation and acknowledges diverse life experiences (de Vries 2019).

In line with this, among the reviewed articles, despite the absence of a direct link to SD and CE in biogas investigation within the Global South, a few of them, including Ahmad and Jabeen (2023), associate biogas benefits with SDGs. They noted biogas' impact on SDG 3—good health and well-being, and SDG 5—gender equality by reducing energy costs and firewood collection time. They also link biogas to SDG 8—decent work and economic growth, and macro welfare benefits to SDG 1—no poverty, and SDG 2—zero hunger. Biogas directly contributes to SDG 7—affordable and clean energy, SDG 9—industry, innovation, and infrastructure, SDG 12—responsible consumption and production, SDG 13—climate action, and SDG 15—life on earth. Mukisa et al. (2022) added SDG 6—clean water and sanitation due to biogas's role in waste management and its direct and indirect contributions to other SDGs like 2, 3, 5, 6, 8, 9, and 13. However, they did not confirm biogas contribution to SDGs 1, 12, and 15.

In addition, Obaideen et al. (2022), one of the review studies that concentrates specifically on the interrelationship between biogas and SDGs, identified the direct contribution of biogas to 12 out of 17 SDGs. They proposed 58 indicators intended to guide and shape stakeholders' engagement and efforts in utilizing biogas for the realization of SDGs. Similarly, Lohani et al. (2021) emphasized the significance of examining the contribution of biogas to SDGs in relation to the size of biogas plants. According to their findings, small-scale biogas plants contribute to various SDGs, including SDG 1—no poverty by enhancing agricultural productivity, SDG 3—good health and well-being by diminishing health incidences, SDG 5—gender equality by empowering girls and women, SDG 7—affordable and clean energy by providing clean fuels, SDG 13—climate action by mitigating climate change, and SDG 15—life on land by forest conservation.

In contrast, Archana et al. (2024) identified inherent challenges associated with the utilization of biogas, irrespective of its potential to advance the fulfillment of the SDGs. Nevzorova and Kutcherov (2019) further elaborated on these challenges by categorizing them regionally into two groups: barriers for the Global South (developing countries) and the Global North (developed countries), while acknowledging some shared issues. Currently, the hurdles confronting developing countries in the adoption of biogas technologies are more significant. Consistent with this observation, the delay in the dissemination of biogas, which impedes its contribution to the SDGs, is not an isolated occurrence. Generally, as articulated by Schröder and Barrie (2024) and Bhandari (2019), a substantial portion of the SDGs is unlikely to be attained by 2030. Only 17% of SDG targets are advancing satisfactorily on a global scale to be achieved by that year, whereas the remaining 83% manifest minimal, stagnant, or regressive progress. At the global level, environment-related goals associated with the planet, especially the energy sector that needs to produce renewable and affordable energy, are the major ones falling behind (Asif et al. 2023). Changing this situation requires changing the current squeezed share of 'green financing' (Sachs et al. 2019). Consequently, rather than abandoning the SDGs, key policymakers are advocating for a reassessment of the current targets and an extension of the SDG timeframe until 2050.

4.2.2 | Trends of CE as a Theme

The results of this systematic scoping review through the manual interpretative coding indicate that CE has the least thematic representation compared to the integrated theme of SD and CE as a distinct theme alongside SD. This is due to the limited evidence and attributes that hinder the establishment of a direct correlation with CE through its two specific subthemes, namely waste-to-energy and waste management, as examined in the reviewed articles. Contrarily, Tagne et al. (2021) explicitly state, based on their findings, that biogas in numerous developing countries has initiated a new scenario as a mechanism to alleviate poverty within a broader circular economic framework.

Historically, the concept of CE is entrenched in environmental economics, delineated through four environmental functions: amenities, inputs for economic activities, waste absorption, and a life support system (Andersen 2007). It is established on

three core principles. Firstly, the regulation of finite resources, alongside the equilibrium in utilizing renewable resources, aims to avert waste and pollution. Secondly, it advocates for the enhancement of resource utility through their perpetual recirculation within their life cycle, thereby forming a closed-loop system. Lastly, it emphasizes improving system efficiency to rejuvenate natural ecosystems (Macarthur Foundation 2024). Ghazanfari (2023) and Ghisellini et al. (2016) support this by asserting that CE is a process that eliminates the unidirectional flow of resources that leads to depletion. Instead, it fosters a closed-loop flow of resources to augment their value and mitigate environmental degradation.

Subsequently, notwithstanding the aforementioned context, the bibliometric analysis of this study did not display any significant variation from the manual interpretative coding analysis. It verifies the infrequency of CE as a theme within the reviewed literature and did not emerge as either an evolving or declining theme concerning its relevance and extent of development over time. Nonetheless, biogas utilizes organic waste as a valuable economic resource with considerable potential to be converted into renewable and green energy through anaerobic digestion (Armah et al. 2019; Djimtoingar et al. 2022). This notion is further reinforced by Paes et al. (2019), who comment on organic waste as a foundation for developing a novel circular economic model capable of generating new energy sources, fostering employment opportunities, reducing expenses, accumulating assets, and enhancing environmental protection. Furthermore, El Ibrahimy et al. (2022b) and Oyelaran et al. (2018) asserted that in the construction of a circular economy utilizing organic waste, biogas technology serves as a viable, environmentally sustainable, and economically beneficial waste management mechanism that can additionally produce organic fertilizers.

Thus, this demonstrates the integration of waste, water, energy, and food systems in delivering environmental benefits (Lin et al. 2021). Biogas plays a role in enabling the recirculation of resources to enhance their value and diminish environmental impacts, thereby creating a closed-loop system (Ghazanfari 2023; Ghisellini et al. 2016). This is exemplified by the findings of Shaibur et al. (2021), which suggest that biogas functions as a crucial method in the circular economy by transforming cow manure and agricultural waste into energy and organic fertilizer that recycles nutrients, thus replacing synthetic fertilizers. The process has produced economic advantages by allowing households to diminish their spending on chemical fertilizers while concurrently achieving improved plant growth (Czekala et al. 2022; Shaibur et al. 2021). Additionally, biogas digestate within the circular bioeconomy context can also be applied for 'animal feed, obtaining irrigation water, bio-pesticide, seed pre-treatment, phosphate salt, and carbon synthesis' (Baştıbak and Koçar 2020).

For numerous nations within the Global South, the predominant impediment to the dissemination of biogas technology to foster a circular economy is the lack of a comprehensive policy to direct all initiatives (Patinvoh and Taherzadeh 2019). As suggested by one of the reviewed articles, Aggarwal et al. (2021), India has observed inadequacies in national biogas programs, underscoring the urgent need for an overarching policy. In

acknowledging this issue, India, akin to China, has drawn lessons from past experiences, subsequently implementing contemporary policies and emerging as the largest consumer of biogas, whereas Europe boasts the highest rate of promotion (Abanades et al. 2022). To date, achieving the objectives of a CE necessitates both expansion and macro-coordination, as it cannot be realized independently by a single nation. Nevertheless, in practical terms, there is a significant divergence from this narrative, characterized by limited objectives, minor and temporary projects, all executed without integration (Schröder and Barrie 2024).

Therefore, in relation to the planning and execution of biogas policies, projects, or programs, as well as other CE-related initiatives, it is essential to consider the three distinctive, scientifically validated levels of functional spheres inherent in the CE framework. Firstly, the micro level signifies the minimal level of implementation achievable within individual industries, production processes, or service delivery entities. The meso level, in contrast, represents the coordinated sphere of interaction among industries, companies, and consumers. Finally, the macro level denotes the most extensive sphere, encompassing numerous stakeholders, and is representative of society at large (Campbell-Johnston et al. 2020; Suárez-Eiroa et al. 2019). Furthermore, across all three functional spheres, the implementation of value retention/R strategies is crucial for the establishment of a CE system. While there is no consensus on their exact number, these strategies include redesigning, reducing, (re)using, reselling, remanufacturing/refurbishing, recycling, recovering, and recirculating (Campbell-Johnston et al. 2020; Garcia-Saravia Ortiz-de-Montellano et al. 2022). These strategies concentrate on the product itself, its components, the materials constituting the product, and the logistics required for product circulation (Garcia-Saravia Ortiz-de-Montellano et al. 2022).

4.2.3 | Trends of the Combination of SD and CE as a Distinct Theme

The findings of this study revealed that the principal theme integrates SD, and the CE as a distinct theme dominates the thematic coverage of the reviewed articles. This is because of the implicit view about biogas as a combined solution to the prevalent traditional waste, herds, and forest management practices. Its far-reaching benefits extend to the production of renewable energy and organic fertilizer, protecting the environment, preventing the release of greenhouse gases, improving living conditions, and creating economic opportunities (Shaibur et al. 2021; Singh and Kalamdhad 2022). In support of this, Dhungana et al. (2022) stated that intensifying the utilization of renewable and green energy that binds all the long list of benefits of biogas within the framework of a circular economy plays a pivotal role in nations' efforts to ensure SD and achieve the SDGs within their time frame. Nevertheless, Ali et al. (2020) and Rupf et al. (2017) stressed the absence of evidence on the effectiveness and appropriateness of implemented biogas plant designs in line with the local social, economic, and environmental conditions to maximize these benefits.

On the other hand, as demonstrated in the results of this study, among the six distinct subthemes under this principal

theme, the environmental subtheme attains the most comprehensive coverage, followed by the waste-to-energy subtheme. Likewise, Afotey and Sarpong (2023) note that the environmental benefit of biogas is prominent in reducing greenhouse gases. Furthermore, Slorach et al. (2019) and Swinbourn et al. (2024) highlight biogas' role in decarbonization through converting waste to biogas, reducing fossil fuel reliance, and mitigating CO₂ emissions. However, Werkneh (2022) points to limitations like impurities, such as carbon dioxide (CO₂), hydrogen sulfide (H₂S), siloxanes, nitrogen oxides (NO_x), ammonia (NH₃), and halogens, that need careful management. Despite this, biogas's environmental benefits are significant, supporting SDGs and aiding the shift to a low-carbon CE (Singh and Kalamdhad 2022; United Nations Environment Programme 2019). In agreement with this, Afridi et al. (2023) highlight China's ambitious challenge to adopt technological innovations and renewable energy, including biogas, to align emission reductions with the global temperature increase limit of 1.5°C and below 2°C. Furthermore, this is supported by innovations in biogas purification, including amine scrubbers for high purity, water scrubbers for cost-effective small-scale use, membrane separation for low methane slip, and promising cryogenic separation for methane (CH₄)-carbon dioxide (CO₂) separation are advancing the technology (Bauer et al. 2013).

Subsequently, this review's findings reveal that the waste-to-energy subtheme received the second highest level of coverage in the reviewed research articles. This is not such an exceptional thing to happen, as biogas production exemplifies a waste-to-energy system. It converts organic waste, including agricultural by-products and municipal waste, offering a renewable energy alternative usable for electricity, heat, and fuel (Wu and Xu 2024; Kulkarni 2016; Wu and Xu 2024). Whereas in the case of the waste management subtheme, it takes third place in our findings. This signifies that the potential of biogas to transform waste handling by reducing landfill and producing digestate as fertilizer (Kulkarni 2016; Wu and Xu 2024). As indicated in the findings of this study, the remaining subthemes, including the economic, social, and technical/technological subthemes, occupy the subsequent places from fourth to sixth, respectively. This suggests, as Ani et al. (2024) stated, that the variation of biogas' economic benefits between regions results from the availability of organic wastes and supportive policies. Likewise, the social subtheme connection with cultural reluctance, like using human waste (Boyd Williams et al. 2022), potentially hampers biogas' prominence within the reviewed studies. In the same way, the technical/technological subtheme encompasses the complexity, the need for skilled workers, and advanced equipment for anaerobic digestion (Sime 2020) contributes to the low level of coverage in the reviewed article.

In brief, notwithstanding the identified benefits and drawbacks that influence the extent of coverage for each subtheme in the reviewed research articles, the present study's findings reveal that the examination of biogas in the context of CE and SD will persist. Ani et al. (2024) and Richards and Yabar (2022) confirm that biogas technology is poised to play a crucial role in the transition to sustainable energy by effectively addressing energy poverty, advancing the SDGs, and providing farmers with supplementary income and reduced

waste management costs through innovation and scientific analysis.

4.3 | Global South Nations' Scientific Contributions on Biogas Related to CE and SD and Overall Regional Contexts

The term Global South Nations traces back its origins to the 1955 Asian–African Bandung Conference. It is also linked with the Non-Aligned Movement (NAM) and the United Nations Conference on Trade and Development (UNCTAD). Its prominence waned with the advent of organizations like the World Intellectual Property Organization, the General Agreement on Tariffs and Trade (GATT), and the World Trade Organization (WTO). Additionally, the economic rise of China and the BRICS nations has considerably influenced it. Since the 1950s, this concept has undergone numerous transformations, encompassing periods of progress and decline (Gray and Gills 2016). The region is known for widespread poverty in different sectors (Kowalski 2021). According to Ebhota and Tabakov (2024), in 2019 alone, despite the Global South's preceding decade's success in over one billion individuals' access to electricity, the region accounted for the top 20 countries on the energy deficit list. This list comprises six nations from Eastern and South-Eastern Asia, ten from Sub-Saharan Africa (SSA), and four from Central and Southern Asia. Thus, under this section, these facts and their extensions are expounded in line with the findings of this study.

4.3.1 | Global South Nations' Scientific Contribution on Biogas Related to CE and SD and Publication Over the Years

Studies on biogas have been and will continue to be undertaken globally. These efforts encompass research that concentrates on the trajectory towards socioeconomic dimensions and the pressing necessity for technological evolution (Apfel et al. 2021). In affirming this, the results of this study indicated that 31 leading nations authored 43 peer-reviewed articles on the nexus of biogas, CE, and SD, whilst a total of 60 nations participated in varying degrees in international collaborative efforts. An exemplary illustration of this is the scrutiny of biogas by Igbum et al. (2019), Kilucha et al. (2022), and Li, Gan, et al. (2023) pertaining to climate change. Their research indicated that increased demand for affordable energy, such as biogas, resulting from population growth, subsequently contributed to the restoration of ecosystems and natural resources. Consequently, appropriate biogas production in Global South Nations, with adherence to all necessary safety standards, has the potential to mitigate greenhouse gas emissions (Afotey and Sarpong 2023; Lemma et al. 2021). In the case of Ethiopia, each biogas-holder household has the potential to reduce 6024 kg CO₂-eqyr.⁻¹ of greenhouse gas emissions (Lemma et al. 2021). In China in the year 2020, 46,794.90 Gg CO₂-eqyr.⁻¹ of greenhouse gas was estimated to be reduced by producing 38.5 billion m³ of biogas (Yu et al. 2008). Whereas in Nepal, from livestock manure alone, 3043.58 million m³ year⁻¹ of biogas production is estimated to potentially prevent 4.35 million tons of CO₂-eqyear⁻¹ greenhouse gas emissions (Lohani et al. 2021). Beyond that, the aggregate result of all these efforts is going to be revealed in the power

of biogas in creating a circular economy and contributing to SD (Kougias and Angelidaki 2018).

On the other hand, according to the findings of this study, among the 31 prominent nations contributing to 43 articles, 21 represent countries from the Global South, while 10 represent developed nations. In contrast, Bhattarai et al. (2022) note that developed countries contribute more to renewable energy research without a regional focus. In agreement with this, Salager-Meyer (2008) highlights that Global North Nations dominate scientific publications due to their industrial advantages and expert retention. Zhou and Leydesdorff (2006) add that the publication gap between regions is due to competitive publishers, a misalignment with national research needs, and limited resources in the Global South, causing global imbalance and poor collaboration. On top of that, language barriers, infrastructure issues, and financial disparities worsen the situation. In response, this creates an immense gap in the effectiveness of Global South Nations governments and international development agencies/organizations' attempts to navigate, guide, formulate, and implement conducive bioenergy/biogas policies, regulations, and programmes/projects that link biogas technology with SD and CE (Chen et al. 2023; Dunmade 2019). Therefore, beyond the recognized challenges that hinder the scientific contributions of Global South Nations, as noted by Surendra et al. (2014), within the Global South, where approximately 3 billion individuals depend on traditional energy sources primarily obtained from the combustion of biomass, the engagement of local scholars in scientific discourse that can lay a ground for the formulation of conducive policy and programs is indispensable.

Concurrently, increasing scientific contributions of the Global South Nations means accelerating the use of the surplus biomass in these regions (Kay Lup et al. 2023). This holds significant potential for biogas production (Li, Gan, et al. 2023) and requires comprehensive initiatives rooted in a bio-circular approach to generate benefits at a community level (Kapoor et al. 2020). Consequently, the outcome would be the development of community-owned biogas technology capable of alleviating social, ecological, financial, and health challenges associated with biomass combustion (Li, Gan, et al. 2023; Mengistu et al. 2015). Furthermore, foreseeable and severe events such as flooding, droughts, and rising sea levels, which critically affect Global South Nations, could be mitigated (Taconet et al. 2020; Thomas et al. 2020). However, the on-ground facts are quite contrary. For instance, in Indonesia, the dissemination of biogas technology is primarily hindered by ineffective governance, compounded by the substantial subsidy on LPG (Budiman 2021). Whereas, in Nepal, despite the widespread trend for the adoption of toilet-linked biogas digesters, technical challenges are found to be the main barrier, surpassing sociocultural norms (Boyd Williams et al. 2022). In alignment with this, in the case of India, an analytical hierarchy process examining the adoption of large-scale biogas plants identified that competition from alternative fuels, the influence of subsidies, insufficient financing mechanisms, and variations in feedstock represent the predominant barriers (Singh and Kalamdhad 2024).

The result of this study regarding the scientific contributions of the individual Global South Nations revealed that China,

which was less active until about 2017, then showed a rapid rise in 2022, occupies the first place. This finding corroborates Bhattacharya et al.'s (2015) finding, which emphasizes China's significant strides across all disciplines in global scientific publication trends. However, despite China's impressive growth in the output of scientific publications that places it in the fifth position globally, it does not stand alone as the only nation from the Global South to achieve such prominence. India, South Korea, and Brazil are also key contributors on this list. The rise of these accomplishments is attributed to an increasingly supportive environment characterized by public financing, a large pool of skilled personnel, and sustained economic progression in these countries (Zhou and Leydesdorff 2006). However, in contrast, the findings of this study bring forth South Africa, in the second place after China, which started early and has consistently maintained a stable, gradually ascending trajectory. Nigeria, Ethiopia, and Mauritania hold third, fourth, and fifth places, respectively, showing moderate upward trends. In perspective of these findings, as Kalina et al. (2022) articulated, it is imperative to emphasize that existing scientific studies on biogas technology insufficiently address the pragmatic issues concerning the delaying factors of the anticipated biogas revolution and project deficiencies that could contribute to the creation of CE and SD. Consequently, it necessitates a comprehensive scientific inquiry, particularly from social scientists, and the active participation of stakeholders within the biogas sector.

Equally, this study's bibliometric analysis results elucidate the scientific impacts of the Global South by analyzing citation metrics. South Africa is at the forefront, followed by Pakistan, Cameroon, Ethiopia, and China. These standings diverge from a generalized aggregate ranking of nations' citations, as revealed by King (2004) and cited by Salager-Meyer (2008). Citation counts serve as indicators of a nation's development and technological status. The USA, UK, Germany, Japan, Canada, France, and Italy are preeminent in citations ranking among a cohort of 31 countries, collectively accounting for 98% of all citations globally among 191 countries. Notably, only China, India, and Iran from the Global South feature among these 31 leading countries. Significantly, there are several individual exceptions, such as in Egypt's information systems sector, where 13% of researchers surpass their peers, contributing to half of the country's publications and a doubling of citation counts (Albanna et al. 2021). Although collaboration among Global South Nations, our findings identified China as a prominent player, followed by South Africa and Pakistan, which contribute comparably. Conversely, Bangladesh and Ethiopia are situated at the lower end of the hierarchy. In particular, Confraria et al. (2017) indicate that collaboration between countries in the publication of scientific papers is affected by prior citations and correlated with their publication volume within a specific scientific discipline. Then this leads us to the exploration of the Global South Nations' overall regional circumstances.

4.3.2 | Overall Regional Contexts of the Global South

Research on biogas technology is primarily conducted in the Global North, while its implementation is comparatively extensively observed in the Global South (Tavera-Ruiz et al. 2023). Contrary to this, despite biogas being one of the most effective

waste management solutions, a considerable number of developing countries face deficiencies in waste management practices and strategies, exacerbated by demographic expansion and economic constraints. Conversely, the Global North enjoys sophisticated and operationally advanced waste management solutions, including the application of biogas (Khan et al. 2022). In line with this, at the national level, numerous Global South Nations often overlook the potential of their bioresources for biogas production by prioritizing solar and wind energy (Afridi et al. 2023). While the focus on solar and wind energy, supported by requisite feasibility studies, is commendable, the neglect of bioresource-based energy sources, such as biogas, significantly impacts their ability to effectively coordinate and integrate CE objectives (Brown 2018).

Conversely, expeditious rectification of this oversight will facilitate the incorporation of renewable energy into a CE, which is crucial for SD (Klemeš et al. 2023). In this regard, like it has been proven to be a key factor for success in Nepal and China, the rest of the Global South Nations need to reinforce their biogas policies and programs by establishing and making accessible financial institutions of their preference for a smooth flow of funds between government institutions, donor agencies, and farmers/end-users (Afridi et al. 2023). They can furnish the necessary capital and expertise required for biogas projects across varying scales, from small to large. The development of public-private partnerships is also essential (Patinvoh and Taherzadeh 2019). Furthermore, as Farrukh et al. (2024) suggested, the involvement of local governance and the engagement of the community are critical in fostering supportive environments. The inclusion of stakeholders such as farmers and businesses ensures that policies adequately address regional needs, thereby enhancing their efficacy. Evidence from developing countries indicates that community involvement significantly augments the acceptance and utilization of biogas systems, a strategy that can be implemented in nations such as Nigeria (Farrukh et al. 2024; Nwankwo et al. 2024).

In some cases, stakeholders might not possess the necessary knowledge to effectively promote biogas technology. This lack of knowledge can result in resistance from organizations like farmer unions, inhibiting implementation efforts. Stakeholders may also harbour concerns about potential project failures, fostering a negative outlook due to unresolved issues, reliance on a narrow range of feedstock, and insufficient policy backing for major financiers (Farrukh et al. 2024). Additionally, as seen in Brandenburg, Germany, the establishment of biogas facilities can lead to disputes over land use because of pre-existing land activities and concerns about the visual impact on the landscape. In these situations, managing conflicts requires robust public negotiation, communication, and designing a biogas system that is both professionally managed and contextually aware, while also engaging all relevant stakeholders (Fienitz 2025). Conversely, in Sogwala village, Zimbabwe (Dhliwayo et al. 2020), research shows that biogas projects can enhance social development by building trust and strengthening social networks. These projects play a crucial role in reinforcing community bonds and networks, laying the groundwork for future developmental projects.

At the regional level within the Global South, distinct challenges are present. In Africa, where 60% of the population lacks access

to reliable energy, the adoption of biogas technology is impeded by market volatility, insufficient awareness, inadequate policy frameworks, and prohibitive costs (Gbadeyan et al. 2024; Ketuama et al. 2022). These barriers undermine financial confidence and restrict access to funding, further exacerbated by high interest rates and unsuitable digester designs (Gbadeyan et al. 2024). Nevertheless, biogas has been shown to enhance living conditions across economic, social, environmental, and technical spheres (Balgah et al. 2023). Similarly, Asia confronts financial challenges in biogas development, with existing policies favoring solar and wind energy, but China and Nepal reap the benefits of well-articulated biogas policies. In Pakistan, various challenges persist despite the availability of abundant resources (Afridi et al. 2023). In South America, the Human Development Index signifies energy options, with Bolivia and Venezuela exhibiting moderate levels, whereas countries like Brazil, Paraguay, and Peru achieve higher benefits from the harnessing of hydroelectric power (Magne et al. 2024). However, the dairy industry in the region necessitates the establishment of effective waste management strategies, like biogas technology, to produce energy and organic fertilizers (Villarroel-Schneider et al. 2022). Overall, Boyd Williams et al. (2024) advocate for a transition to a justice-centered research and development framework in the Global South biogas sector to enhance stakeholder accountability.

Even though the finding of this study listed the term biogas, renewable energy, biogas potential, anaerobic digestion, and co-digestion as key keywords, Laramee and Davis (2013), conversely indicated that all biogas-favoring features, conditions, a firsthand research work on the dissemination of biogas, and its contribution to SD are very low in the Global South Nations. In exception to this, as stated by Li, Gan, et al. (2023) and Li, Wan, et al. (2023) and Zhang and Qiu (2018), China has the highest installation number among the developing nations. They also link biogas with their agroecology by initiating a 'centralized biogas supply system' to support the country's goal of reaching the carbon emission peak by 2030 and carbon neutral by 2060 in building a sustainable economy. In general, as Tigabu et al. (2015) articulated, the reason for the less favorable conditions for the dissemination of biogas in the Global South Nations is attributed to the Global South Nations' priority on establishing and up-keeping a modern and clean household energy infrastructure, isolated from the comprehensive framework of SD.

Overall, concerning SDGs, as Bhandari (2019) concludes that no nation, including the Global South, is currently progressing in a manner that ensures its attainment. As Schröder and Barrie (2024) stated, accomplishing objectives and benchmarks within SD models requires the utilization of suitable mechanisms at pivotal leverage points, facilitating the transformation of regulatory, institutional, financial, and various communal frameworks. Meanwhile, the Global South Nations' ongoing efforts to achieve SD by fulfilling the SDGs are lagging and becoming more difficult due to their economic, political, and technological limitations (Wei et al. 2023). Yet, they are under unique pressure to push further toward fulfilling SDGs, enclosed by the existing imbalances in global resource distribution and theft in their domestic institutional functioning (Medina-Hernández et al. 2023; Wei et al. 2023). If measures are not taken swiftly to avert the enclosure, it will continue to greatly

impact the Global South Nations' effort to tackle other economic and social-related goals (Ebhotu and Tabakov 2021; Kay Lup et al. 2023; Wei et al. 2023). In averting this reality, biogas is an instrumental, promising technology as an alternative source of renewable energy and nutrient recycling (Kougias and Angelidaki 2018; Lohani et al. 2021).

5 | Conclusions and Recommendations

5.1 | Conclusion

Initially, this study sought to examine the trends and gaps in methodological approaches and thematic recurrence within selected peer-reviewed research articles. To fulfill this objective, the study implements a systematic scoping review, creating a table based on a developed manual interpretative coding framework, followed by a bibliometric analysis utilizing RStudio's Biblioshiny interface. Within this context, our findings reveal that peer-reviewed scientific articles explore their research inquiries through a variety of sophisticated methodologies. Among these methodologies, there is a notable predominance of the quantitative method with an experimental design. Regarding thematic recurrence, the manual interpretative coding analysis suggests a relatively inconsistent incorporation of CE and SD frameworks in the investigation of biogas. In most instances, biogas is studied as a technological solution that can improve access to clean and affordable energy in the Global South Nations. Nonetheless, the bibliometric analysis did not indicate any significant trend of investigating biogas using the CE and SD frameworks. Ultimately, this review elucidates and substantiates the instrumental role of biogas in the effective establishment of the CE and SD.

These results suggest the necessity to balance the interplay of various research methodologies in different studies to preserve and comprehend the flexibility of the conducted analyses, the efficacy in estimation and forecasting, and to attain credible scientific rigor. This will significantly resolve the accumulation of studies on experimental manipulation of anaerobic digestion and feedstock valorization. Similarly, the investigation of biogas needs to primarily emphasize its association with the CE and SD frameworks beyond the assessment of anaerobic digestion as a means to generate renewable and green energy, and the way to manipulate it. In doing so, the complementary relationship between the SDGs and the CE requires a wise investigative approach to the SDGs that includes circular solutions to accelerate and reinforce mutual benefits. As we have surpassed the 9th year of Agenda 2030, with only around 5 years remaining, a particular emphasis is given to the SDGs. Therefore, it is imperative to explicitly examine the link and contribution of biogas to SD, the circular economy, and the SDGs. This inquest is necessary to advance from the current limited state of effort, evidenced by few peer-reviewed studies and a small number of review articles. In conclusion, the role and status of Global South Nations in the realm of global scientific publications, particularly concerning small-scale biogas plants in the framework of CE and SD, warrant encouragement and support by any available means due to the technology's alignment with sustainable solutions for energy scarcity.

To date, investigations into biogas have persisted from various perspectives and appear to be approaching a pivotal stage as global awareness of environmental concerns, such as climate change and global warming, increases due to the conventional reliance on a non-renewable energy-depleting linear economic development model. Conversely, this underscores the necessity of analyzing biogas in conjunction with CE, SD, and the SDGs to address the aforementioned challenges. In this context, this study is distinguished by its unique approach, intensity, magnitude, and focal point, which collectively seek to highlight the significance of integrating biogas with CE and SD over an extensive timeframe. Nonetheless, additional research is required to substantiate its findings within the specific framework of the SDGs.

5.2 | Recommendations

The widespread application of quantitative research methodologies underscores the imperative need for a more balanced research method that integrates both qualitative and quantitative research methods. Subsequently, in facilitating the evaluation of policy alternatives and directions, as well as the integration of IT products and IoT, a balanced approach is indispensable. This should underpin the crucial role of SD in addressing contemporary complex challenges, including global environmental degradation, economic inequality, and social injustice. Consequently, given the assertion that biogas contributes to the enhancement of economic, social, environmental, and technological conditions within a society, it is essential for scientific research to explicitly and comprehensively examine biogas in relation to SD. This will significantly transform the current fragmented efforts based on the perception of a singular SD dimension into deliberate, evidence-driven, and pertinent scientific inquiries for the Global South Nations. Most importantly, in order to fully explore the latent, multifaceted, and interconnected advantages of biogas technology in achieving SD, the adoption of a balanced methodological approach is crucial.

Therefore, for the Global South Nations to develop effective channels for the exchange and transfer of knowledge, experience, and resources tailored to their specific contexts and requirements, it is highly advisable to promote and underscore the significance of collaboration among themselves and particularly with the Global North Nations. Such cooperation will play a pivotal role in enhancing scientific outputs that effectively address their needs. This study ultimately advocates for continued realistic efforts towards strong cooperation. Hence, governmental and non-governmental organizations within the biogas sector must align with this approach in exploring the potential and applicability of technology to address the prevailing energy deficit in the Global South Nations.

In future research endeavors, it is imperative to utilize additional or alternative databases and gray literature, thereby ensuring a meticulous search for credible research articles by adhering to a stringent methodological framework. Furthermore, the inclusion of pivotal peer-reviewed articles, which may be restricted by paywalls or authored in other languages, where feasible, is vital to enhancing the authenticity of the findings. It is also prudent to employ varied and alternative keywords to optimize

the inclusion of peer-reviewed research articles pertinent to the field. Additionally, we recommend the organization of studies within a thematically structured reporting framework to enhance the comprehensive thematic analysis of this topic, all the while upholding the integrity of quality and scientific precision.

Furthermore, upcoming research should purposefully contest the existing status quo, which predominantly concentrates on manipulating the physicochemical and biological elements of the anaerobic digestion process. Hence, it is crucial to broaden the scope of investigation to evaluate the efficacy and appropriateness of biogas technology and plant designs to local social, economic, and environmental conditions that facilitate the realization of CE and SD. In addition, further study needs to focus on understanding the role, contribution, and link of biogas with the SDGs. Such an investigative approach will address the substantial gap encountered by the governments of Global South Nations and international development organizations in formulating favorable biogas/bioenergy policies and programs that integrate biogas technology with CE, SD, and the SDGs.

5.2.1 | Policy Recommendations

Policies on biogas solutions can be established at multiple tiers, encompassing both international and local levels. Thus, on a global scale, policymakers should reflect upon the incorporation of the findings of this study, which necessitates the promotion of inclusive circular models to facilitate the exchange of technology that promotes CE and SD. This establishes standardized measures and metrics for CE and biogas technologies. This is crucial for the execution of globally conceived policy frameworks such as the SDGs and CE, particularly in the context of advancing and disseminating biogas technologies. These will mostly benefit the Global South Nations. On the other hand, at the national level, biogas/bioenergy policies should emphasize the importance of collaboration among governments, businesses, communities, and individuals to create sustainable practices that promote the dissemination of biogas technology. Beyond this, clear ways and mechanisms for cooperation between nations, especially between the Global South and North, should be indicated in the policy contexts.

Equally, it is crucial to coordinate diverse responsibilities and initiatives on the governance of biogas/bioenergy policies by delineating clear responsibilities and obligations to the pertinent sectoral government offices and policy domains. This strategy should also identify where collaborative efforts are necessary. Concurrently, policies should be grounded in local resources and direct the integration of biogas with agriculture, climate change mitigation, waste management (including wastewater treatment plants), and environmental protection.

Overall, policies should be devised in line with CE and SD frameworks to create adaptable policies tailored to the socio-economic contexts of Global South Nations. Hence, policies should explicitly articulate long-term agile strategies and policy instruments to promote technological adoption. These initiatives may include the incorporation and integration of upcycling and recycling concepts of the CE across all sectoral domains, thereby facilitating the enhancement, preservation, and optimization

of the constructive use of natural resources. Moreover, such policy directions are likely to spur innovation that can result in enhanced process efficiency. When it comes to policy instruments, these may include investment support for establishing new plants through governmental subsidies or coordinated loans from financial institutions that promote the utilization of accessible and surplus feedstocks (organic waste). These policies should incentivize research and development of biogas production to fulfill the specific country's prioritized needs, such as household cooking, heat and/or electricity generation, or road transport. Finally, a bottom-up approach is advised, adapting implementation to the local environment. This ensures insights for achieving policy goals and maintains stakeholder support. Key elements like measurement, evaluation, learning, feedback, and knowledge sharing should start at the grassroots. A framework for feedback at any policy stage allows flexible, iterative integration of perspectives and necessary revisions.

Author Contributions

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All materials used for this systematic scoping article are available in the public domain except the protocol and manual interpretative coding analysis, which can be found on request. PRISMA checklist can be found on PRISMA's official website, and articles can be traced back to the respective database.

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Appendix A

Summary tables of a manual interpretative coding analysis classification of principal classification with their aligned specific research design and subthemes with their coverage

Roll No.	Major findings of the study	Total # of articles	Percentage out of the aggregate	Subthemes	No. of articles	Percentage out of the aggregate of subthemes	References
1	Thematic classification	16	37	Economic aspect	12	41	(Ahmad and Jabeen 2023; Ali et al. 2023; Amigun and von Blottnitz 2007; Amigun and Von Blottnitz 2009; Bakehe 2021; Berhe et al. 2017; Chen et al. 2023; Dhlwayo et al. 2020; Laramee and Davis 2013; Nketiah et al. 2022; Nzila et al. 2012; Rasheed et al. 2016; Singh and Kalamdhad 2022; Tembo et al. 2022; Tigabu et al. 2015; Wang et al. 2020)
				Social aspect	7	24	
				Environmental aspect	7	24	
				Technical or Technological aspect	3	10	
2	Methodological classifications	10	23	Waste management	3	23	(Armah et al. 2019; Chukwuma and Orakwe 2014; Djimtoingar et al. 2022; El Ibrahim et al. 2022a; Igbum et al. 2019; Mekonen et al. 2023; Ofefule et al. 2011; Oyelaran et al. 2018; Sánchez et al. 2021; Santos et al. 2022)
				Waste to energy	10	77	
		17	40	Economic aspect	5	11	(Bundhoo and Surroop 2019; Ebeya et al. 2022; El Ibrahim et al. 2022b; Ali et al. 2020; Dunmade 2019; Getahun et al. 2014; Hasan et al. 2022; Kefalew and Lami 2021; Kilucha et al. 2022; Kusmayadi et al. 2023; Li, Gan, et al. 2023; Mboumboue and Njomo 2018; Rasheed et al. 2019; Rupf et al. 2017; Shaibur et al. 2021; Yan et al. 2023; Zhang and Qiu 2018)
				Social aspect	3	6	
				Environmental aspect	15	32	
				Technical/Technological aspect	3	6	
		29	67	Waste management	9	19	(Armah et al. 2019; Chukwuma and Orakwe 2014; Djimtoingar et al. 2022; Ebeya et al. 2022; El Ibrahim et al. 2022a; Getahun et al. 2014; Kilucha et al. 2022; Kusmayadi et al. 2023; Ofefule et al. 2011; Oyelaran et al. 2018; Sánchez et al. 2021; Santos et al. 2022; Amigun and von Blottnitz 2007; Amigun and Von Blottnitz 2009; Yan et al. 2023; Ali et al. 2023; Mekonen et al. 2023; Laramee and Davis 2013; Ahmad and Jabeen 2023; Bakehe 2021; Berhe et al. 2017; Hasan et al. 2022; Nketiah et al. 2022; Wang et al. 2020; M. M. Ali et al. 2020; El Ibrahim et al. 2022b; Li, Gan, et al. 2023; Zhang and Qiu 2018)
				Waste to energy	12	26	
				Experimental design	13	45	
				Survey design	12	41	
		3	7	Case study	4	14	(Dhlwayo et al. 2020; Tembo et al. 2022; Tigabu et al. 2015)
				Project evaluation	2	67	
				Process analysis	1	33	
				Life cycle assessment	2	18	
	Mixed method	3	7	Environmental impact assessment	1	9	(Bundhoo and Surroop 2019; Mboumboue and Njomo 2018; Singh and Kalamdhad 2022; Kefalew and Lami 2021; Chen et al. 2023; Shaibur et al. 2021; Rupf et al. 2017; Rasheed et al. 2016; Rasheed et al. 2019; Dunmade 2019; Nzila et al. 2012)
				Case study	2	18	
				Survey	6	55	

Source: Author.